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QUESTION BANK OF AI&DS-II

B.E SEM-VII **SCHEME: R-19 C SCHEME** **FOR WINTER SESSION 2026**

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Q.1) What is the Bayesian Belief Network? Explain Interfacing in detail with example.

Q.2) What is a Markov Decision Process (MDP)? List its primary components and need of MDP in detail.

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Q.3) Explain the architecture and components of CNN in detail and state its application also.

Q.4) Explain different types of membership functions used in fuzzy logic with suitable example. Describe the Properties of Fuzzy Sets with an example.

Q.5) Explain the autoencoder with architecture diagram in detail.

Q.6) Explain the architecture and key components of ANN in details.

Q.7) Define Defuzzification. Discuss any two methods of defuzzification?

Q.8) What are all the objects of using LSTEM? Explain its architecture.

Q.9) What is cognitive computing, and how does it differ from traditional computing? List and explain the process of building a cognitive application.

Q.10) Define Cognitive computing. Draw a neat diagram of components of the cognitive system and explain the components also List and explain the design principle of Cognitive system.

Q.11) Briefly explain Data science for Multimodal applications.

Q.12) How bagging and boosting handle bias-variance trade-off differently, and analyze their effectiveness in dealing with noisy data and overfitting. Explain with algorithmic such as Random

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Q.13) Explain ensemble methods in details.

Q.14) Perform a case study on video recommendation systems. (data science based)

Q.15) Describe Deep Learning concept with an example. State and elaborate the applications of deep learning.

Q.16) Explain various metrics for evaluating the classifier performance. Discuss any two cross validation methods.

Q.17) What is activation function? Discuss different types of activation function in Neural networks.

Q.18) List three industries where data science is extensively used and briefly describe one application in each industry.

Q.19) Discuss the limitation of deterministic approaches and how probability reasoning address these limitation with example.

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Q.20) How does class imbalanced affect on classification? What are the ways to solve class imbalanced problem.

Q.21) Using Mamdani fuzzy model design a fuzzy logic controller to determine the wash time of domestic washing machine. Assume the inputs are dirt and grease on clothes. Use three descriptors for each input variable and five descriptor for output variables. Derive necessary

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Q.21) Using Mamdani fuzzy model design a fuzzy logic controller to determine the wash time of domestic washing machine. Assume the inputs are dirt and grease on clothes. Use three descriptors for each input variable and five descriptor for output variables. Derive necessary membership functions and required fuzzy rules for the application.

Q.22) Explain the role of representing knowledge in taxonomies and ontologies, and how advanced analytics can be applied to cognitive systems.

Q.23) Discuss the role of utility and expected utility in decision theory with example.

Q.23) Differentiate between the following:

i) RNN and ANN
ii) Fuzzyness and probability
iii) Crips set and fuzzy set
iv) Bagging and boosting

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Q.24) Write a short note on:

i) Trends in Data science for audio
ii) Mc-Culloch pits neuron with an example
iii) Describe NLP in support of cognitive system
iv) Bootstrap for sampling
v) Holdout method and Random subsampling
vi) Significance of ROC curves

Numerical

Q.1) Calculate Accuracy, precision, recall, sensitivity and specificity for the following example.

Q.2) Implement AND, XOR function using Mc-Culloch Pitts model.

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Q.1) What is the Bayesian Belief Network? Explain Interfacing in detail with example.

Ans: A Bayesian Belief Network (BBN) is a probabilistic graphical model used to represent uncertain knowledge and relationships between variables. It consists of nodes, directed edges and conditional probability tables (CPTs), which help calculate the probability of events based on available evidence.

1) A node represents a random variable, such as Rain, Disease or Traffic, while a directed edge represents a probabilistic dependency between two variables.
2) A conditional probability table (CPT) specifies the probability of a variable for different combinations of its parent variables. Variables without parent nodes have prior probabilities.
3) A Bayesian Network is represented using a directed acyclic graph (DAG), meaning that its directed edges do not form a cycle. This structure helps represent dependencies and simplify probability calculations.
4) Inference is the process of calculating the probability of a query variable using known evidence. It allows the network to update its predictions when new information becomes available.
5) Inference uses Bayes' theorem to calculate conditional probabilities.

$$P(A \mid B) = [P(B \mid A) \times P(A)] / P(B)$$

Here, P(A) is the prior probability, P(B | A) is the likelihood, and P(A | B) is the posterior probability after observing B.

6) During inference, the query variable and observed evidence are identified. The relevant probabilities are calculated using the network structure and CPTs, and the results are normalized to obtain valid probabilities.
7) Exact inference methods, such as enumeration and variable elimination, calculate probabilities mathematically. Approximate methods, such as sampling, estimate probabilities when exact calculations are computationally expensive.

Example of Bayesian Network Inference

1) **Network Structure:** Consider a simple Bayesian Network that represents the relationship between Rain and Wet Grass. Rain can cause the grass to become wet, so the network contains a directed edge from Rain to Wet Grass.

$$\textbf{Network Structure: } Rain \rightarrow Wet\ Grass$$

2) **Given Probabilities:** Suppose the following probabilities are given:
   - $P(Rain) = 0.2$
   - $P(No\ Rain) = 0.8$

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2) **Given Probabilities:** Suppose the following probabilities are given:
   - $P(Rain) = 0.2$
   - $P(No\ Rain) = 0.8$
   - $P(Wet\ Grass \mid Rain) = 0.9$
   - $P(Wet\ Grass \mid No\ Rain) = 0.1$
3) **Applying Bayes' Theorem:** If the grass is observed to be wet, we can calculate the probability that it rained using Bayes' theorem.

Formula:

$P(Rain \mid Wet\ Grass) = [P(Wet\ Grass \mid Rain) \times P(Rain)] / P(Wet\ Grass)$

First, calculate the probability of wet grass:

$P(Wet\ Grass) = [P(Wet\ Grass \mid Rain) \times P(Rain)] + [P(Wet\ Grass \mid No\ Rain) \times P(No\ Rain)]$

$P(Wet\ Grass) = (0.9 \times 0.2) + (0.1 \times 0.8)$

$P(Wet\ Grass) = 0.18 + 0.08 = 0.26$

Now, calculate the probability of rain given wet grass:

$P(Rain \mid Wet\ Grass) = (0.9 \times 0.2) / 0.26$

$P(Rain \mid Wet\ Grass) = 0.18 / 0.26$

$$\boldsymbol{P(Rain \mid Wet\ Grass) \approx 0.6923 = 69.23\%}$$

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Therefore, the probability that it rained, given that the grass is wet, is approximately **69.23%**.

4) **Applications:** Bayesian Network inference is used in medical diagnosis, weather prediction, fault detection, and risk assessment. It helps make informed predictions when information is uncertain or incomplete.

Q.2) What is a Markov Decision Process (MDP)? List its primary components and need of MDP in detail.

Ans: A Markov Decision Process (MDP) is a mathematical framework used for modelling decision-making problems in which an agent interacts with an environment and receives rewards for its actions. It is widely used in Artificial Intelligence and Reinforcement Learning to

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Primary Components of MDP

1) **States (S):** States represent the different situations in which an agent can exist. For example, in a robot navigation system, each position of the robot can be considered a state.
2) **Actions (A):** Actions are the possible decisions an agent can take in a particular state. For example, a robot may move forward, backward, left or right.
3) **Transition Probability (P):** It represents the probability of moving from one state to another after performing an action. It is written as P(s' | s, a), where s is the current state, a is the action and s' is the next state.
4) **Reward Function (R):** The reward function specifies the immediate reward or penalty received after taking an action or entering a state. It guides the agent towards desirable outcomes.
5) **Policy (π):** A policy defines the strategy used by an agent to select an action in a particular state. The objective is to learn a policy that maximizes the expected cumulative reward.
6) **Discount Factor (γ):** The discount factor determines the importance of future rewards compared with immediate rewards. Its value lies between 0 and 1, where a higher value gives greater importance to future rewards.

Need for Markov Decision Process

1) MDP provides a structured mathematical framework for solving sequential decision-making problems in which the outcome of an action may be uncertain.
2) It helps an agent select suitable actions by considering the current state, possible future states and rewards received from the environment.
3) MDP supports decision-making under uncertainty by representing possible state transitions using probabilities rather than assuming that every action produces a fixed outcome.
4) It helps determine an optimal policy that maximizes the expected cumulative reward over time instead of focusing only on immediate rewards.
5) MDP is useful in Reinforcement Learning algorithms, such as Q-learning and policy iteration, where an agent learns to improve its decisions through interaction with the environment.

Example of MDP

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Example of MDP

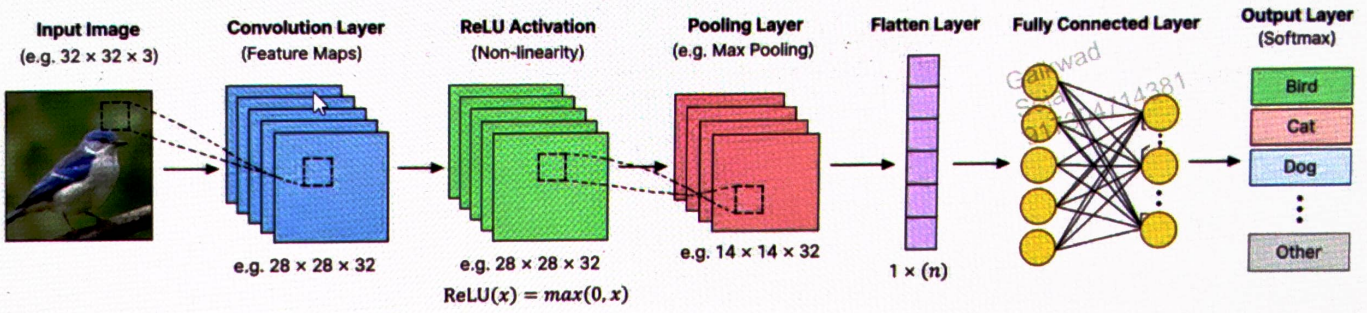
Consider a robot that must reach a destination in a room while avoiding obstacles.

- The robot's position represents the state, and moving left, right, forward or backward represents an action. The transition probabilities describe the possible positions reached after each movement.
- Reaching the destination gives a positive reward, while hitting an obstacle may result in a negative reward. The robot learns a policy that helps it reach the destination while maximizing its expected cumulative reward.

Q.3) Explain the architecture and components of CNN in detail and state its application also.

Ans: A Convolutional Neural Network (CNN) is a type of deep learning neural network mainly used to process images and visual data. It automatically extracts features such as edges, shapes, textures, and objects to perform classification and recognition tasks.

Architecture of CNN



Working of CNN Architecture:

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1) The input image is given to the CNN in the form of pixel values, such as height × width × channels.
2) The convolution layer applies filters to extract features such as edges, corners, textures, and shapes from the image.

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3) The ReLU activation function introduces non-linearity and helps the network learn complex patterns.

$$ReLU(x) = max(0, x)$$

4) The pooling layer reduces the size of feature maps while retaining important information, which reduces computation.

5) The flatten layer converts the feature maps into a one-dimensional vector for further processing.

6) The fully connected layer combines the extracted features and uses them to identify patterns and classify the input.

7) The output layer produces the final prediction. For multi-class classification, the softmax function can calculate the probability of each class.

Components of CNN

1) **Input Layer:** Accepts an image as pixel values, usually represented by height, width, and colour channels.

2) **Convolution Layer:** Uses learnable filters or kernels to extract important visual features from the input image.

3) **Activation Function:** ReLU is commonly used to introduce non-linearity, enabling the network to learn complex relationships.

4) **Pooling Layer:** Reduces the spatial dimensions of feature maps. Max pooling selects the maximum value from a local region.

5) **Flatten Layer:** Converts multidimensional feature maps into a one-dimensional vector.

6) **Fully Connected Layer:** Connects neurons to combine extracted features and perform classification.

7) **Output Layer:** Produces the final class prediction or class probabilities using an appropriate activation function.

Applications of CNN

1) Image classification, such as identifying cats, dogs, and other objects.

2) Object detection in images and videos.

3) Medical image analysis, including the detection of abnormalities in scans.

4) Facial recognition and biometric authentication.

5) Autonomous vehicles for detecting pedestrians, traffic signs, and other vehicles.

6) Handwritten character recognition and optical character recognition (OCR).

7) Video analysis and surveillance systems.

8) Image segmentation for identifying specific regions within an image.

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8) Image segmentation for identifying specific regions within an image.

Q.4) Explain different types of membership functions used in fuzzy logic with suitable example. Describe the Properties of Fuzzy Sets with an example.

Ans: A membership function in fuzzy logic assigns a membership value between 0 and 1 to an element, indicating the degree to which it belongs to a fuzzy set. It is represented as μA(x), where 0 means no membership and 1 means full membership.

Types of Membership Functions

1) **Triangular Membership Function:** It has a triangular shape defined by three parameters (a, b, c). The membership value increases from 0 to 1 and then decreases to 0.

2) **Trapezoidal Membership Function:** It has a trapezoidal shape defined by four parameters (a, b, c, d). The membership value increases gradually, remains at 1 over an interval, and then decreases.

3) **Gaussian Membership Function:** It has a smooth, bell-shaped curve defined by its centre and standard deviation. It is useful when gradual transitions between fuzzy categories are required.

4) **Generalized Bell Membership Function:** It has a smooth bell-shaped curve controlled by three parameters that determine its width, slope, and centre.

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5) **Sigmoidal Membership Function:** It has an S-shaped curve and is commonly used to represent concepts that gradually increase or decrease, such as high temperature.

6) **Singleton Membership Function:** It assigns a membership value of 1 at a particular point and 0 elsewhere. It is often used in simplified fuzzy systems.

Example: For the fuzzy set *Warm Temperature*, a temperature of 25°C may have a membership value of 0.7, meaning it belongs to the set Warm to a degree of 0.7.

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Properties of Fuzzy Sets

Consider a fuzzy set A representing *Warm Temperatures*, with membership values:

$$\mu A(20°C) = 0.3, \mu A(25°C) = 0.7, and\ \mu A(30°C) = 1.0.$$

1) **Membership:** Each element has a membership value between 0 and 1, indicating its degree of belonging to the fuzzy set.
2) **Support:** The support contains all elements whose membership values are greater than 0. For set A, these include 20°C, 25°C, and 30°C.
3) **Core:** The core contains elements whose membership value is exactly 1. Here, 30°C belongs to the core.
4) **Boundary:** The boundary contains elements whose membership values are greater than 0 but less than 1. Here, 20°C and 25°C are boundary elements.
5) **Normality:** A fuzzy set is normal if at least one element has a membership value of 1. Set A is normal because μA(30°C) = 1.
6) **Convexity:** A fuzzy set is convex if the membership value between two points is at least the minimum of their membership values. It ensures there are no dips between points in a one-dimensional membership curve.
7) **Equality:** Two fuzzy sets are equal if their membership values are identical for every element in their domain.
8) **Subset:** A fuzzy set A is a subset of B if $\mu A(x) \leq \mu B(x)$ for every element x.
9) **Complement:** The complement of A is defined as $\mu\bar{A}(x) = 1 - \mu A(x)$. For 25°C, the complement membership is 1 − 0.7 = 0.3.
10) **Union and Intersection:** The union takes the maximum membership value, while the intersection takes the minimum membership value for each element. For example, if $\mu A(x) = 0.7\ and\ \mu B(x) = 0.5$, then union = 0.7 and intersection = 0.5.

Q.5) Explain the autoencoder with architecture diagram in detail.

Ans: An **autoencoder** is a type of artificial neural network used for unsupervised learning. It learns to compress input data into a smaller representation and then reconstruct the original data from this compressed representation. It consists of three main components: Encoder, Bottleneck (Latent Space), and Decoder.

Architecture of Autoencoder

The architecture of an autoencoder consists of an input layer, an encoder that compresses the input data, a bottleneck that stores the compressed representation, and a decoder that reconstructs the original input.

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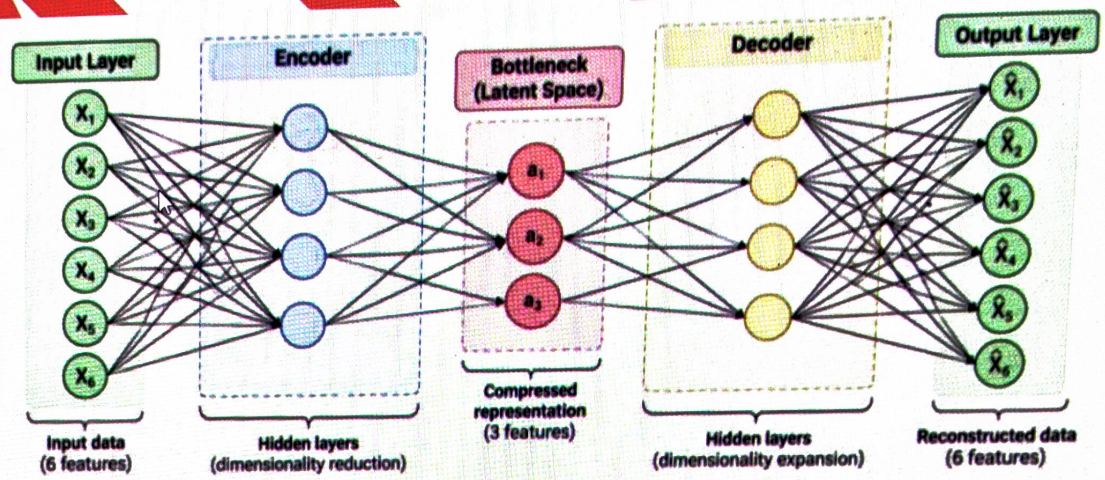
Architecture Diagram:



For example, if the input contains six features ($(X_1, X_2, X_3, X_4, X_5, X_6)$), the encoder may compress them into three latent features ((a_1, a_2, a_3)), and the decoder reconstructs the six output features.

1. Encoder

1) The encoder is the first main component of an autoencoder that receives the original input data.
2) It processes the input through multiple neural network layers to extract important features.
3) Each layer applies weights, biases, and activation functions to transform the input data.
4) It reduces the dimensionality of the input while preserving the most important information.
5) The encoder produces a compressed representation of the input, which is passed to the bottleneck layer.
6) It helps remove redundant information and learn meaningful features from the input data.

2. Bottleneck (Latent Space)

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2. Bottleneck (Latent Space)

1) The bottleneck is the central and usually smallest layer of an autoencoder that stores the compressed representation of the input data.

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2) It is also known as the latent space or latent representation.
3) It forces the network to retain only the most important features of the input data.
4) It reduces the amount of information that needs to be processed by removing unnecessary details and redundancy.
5) The compressed representation is passed from the bottleneck to the decoder for reconstruction.
6) It helps the autoencoder learn efficient representations that can be useful for feature extraction and dimensionality reduction.

3. Decoder

1) The decoder is the final main component of an autoencoder that receives the compressed representation from the bottleneck layer.
2) It processes the encoded information through multiple neural network layers to reconstruct the original input.
3) The decoder gradually expands the compressed representation into a higher-dimensional representation.
4) It uses learned weights, biases, and activation functions to restore the important features of the input.
5) The output layer produces the reconstructed data, which should be as close as possible to the original input.
6) The decoder is trained along with the encoder to minimize the difference between the original input and the reconstructed output.

Working of an Autoencoder

1) The original input data is provided to the encoder.
2) The encoder extracts important features and compresses the input into a smaller representation.
3) The bottleneck stores the compressed representation in the latent space.
4) The decoder uses this representation to reconstruct the original input data.
5) The reconstructed output is compared with the original input using a reconstruction loss function, such as Mean

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Squared Error (MSE).

6) The network updates its weights through backpropagation to minimize the reconstruction loss and improve the quality of the output.

Thus, an autoencoder learns an efficient compressed representation of input data through its encoder and bottleneck, while its decoder reconstructs the input as accurately as possible. It is commonly used for dimensionality reduction, feature extraction, noise removal, and anomaly detection.

Q.6) Explain the architecture and key components of ANN in details.

Ans: An Artificial Neural Network (ANN) is a machine learning model inspired by the structure and functioning of the human brain. It consists of interconnected artificial neurons that process input data, learn patterns, and generate predictions by adjusting their internal parameters during training.

Architecture of ANN

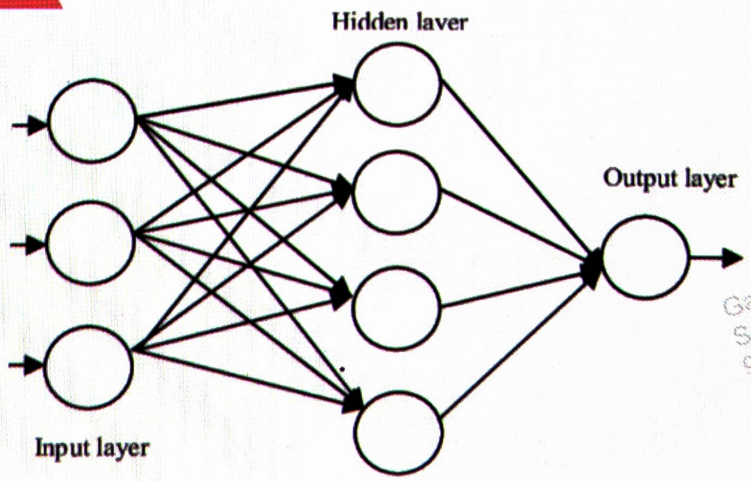
The architecture of ANN consists of three main types of layers:

1) **Input Layer:** The input layer receives the features of the dataset and passes them to the next layer. Each neuron generally represents one input feature, such as age, temperature, or image pixel value.

2) **Hidden Layer:** The hidden layer processes input data using weights, biases, and activation functions. An ANN can contain one or more hidden layers to learn simple as well as complex relationships in the data.

3) **Output Layer:** The output layer produces the final prediction based on the problem being solved. It may generate a numerical value for regression or class probabilities for classification.

ANN Architecture Diagram:



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Key Components of ANN

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1) **Artificial Neurons:** Artificial neurons are the basic processing units of an ANN. Each neuron receives inputs, multiplies them by corresponding weights, adds a bias, and applies an activation function to generate an output.
2) **Weights:** Weights represent the strength and importance of connections between neurons. During training, the network adjusts these values to reduce prediction errors and improve its performance.
3) **Bias:** Bias is an additional learnable parameter added to the weighted sum of inputs. It allows a neuron to shift its activation and learn patterns more effectively, even when the input values are zero.
4) **Activation Function:** An activation function determines the output of a neuron and introduces non-linearity into the network. Common activation functions include ReLU, sigmoid, and tanh, which help the network learn different types of relationships.
5) **Layers:** Neurons are organized into input, hidden, and output layers. Each layer performs a specific role in receiving data, extracting patterns, and producing the final result.
6) **Loss Function:** A loss function measures the difference between the predicted output and the actual output. Examples include Mean Squared Error (MSE) for regression and cross-entropy loss for classification.
7) **Learning Algorithm:** The learning algorithm updates the network's weights and biases to minimize the loss. Backpropagation calculates gradients of the loss, while an optimizer such as gradient descent uses these gradients to update the parameters.

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Working of ANN

1) **Input Processing:** Input data is supplied to the input layer, where each neuron receives a feature value and passes information to the next layer.

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Working of ANN

1) **Input Processing:** Input data is supplied to the input layer, where each neuron receives a feature value and passes information to the next layer.
2) **Weighted Sum Calculation:** Each neuron multiplies its inputs by their corresponding weights, adds the results, and then adds a bias.

$$z = \Sigma(w_i \times x_i) + b$$

Here, w_i represents weights, x_i represents input values, and b represents bias.

3) **Activation:** The activation function processes the weighted sum to produce the neuron's output. This allows the network to learn complex patterns that cannot be represented by simple linear relationships.

Output = f(z)

4) **Prediction and Error Calculation:** The output layer generates a prediction, which is compared with the actual value using a loss function to determine the network's error.
5) **Backpropagation and Weight Update:** Backpropagation calculates the gradients of the loss with respect to the network parameters. The optimizer uses these gradients to update weights and biases, reducing the error over repeated training iterations.

Applications of ANN

1) **Image and Speech Recognition:** ANNs learn patterns in images and audio to identify objects, recognize spoken words, and classify visual information.
2) **Medical Diagnosis:** They analyze medical data to support disease prediction, medical image analysis, and clinical decision-making.
3) **Forecasting:** ANNs learn patterns from historical data to predict weather conditions, sales, demand, and financial trends.
4) **Fraud and Anomaly Detection:** They identify unusual patterns in financial transactions, network activity, and other datasets that may indicate fraud or abnormal behaviour.
5) **Natural Language Processing:** ANNs are used in language translation, text classification, sentiment analysis, and intelligent chatbots.
6) **Recommendation Systems:** They analyze user preferences and behaviour to recommend products, movies, music, and other relevant content.

Q.7) Define Defuzzification. Discuss any two methods of defuzzification?

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Q.7) Define Defuzzification. Discuss any two methods of defuzzification?

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Ans: Defuzzification is the process of converting the fuzzy output obtained from a fuzzy inference system into a single crisp numerical value. It is the final step in a fuzzy logic system and helps convert fuzzy decisions into practical actions.

1. Centroid Method (Centre of Gravity Method)

1) The centroid method calculates the centre of gravity of the aggregated fuzzy output membership function. It is one of the most commonly used defuzzification methods.

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2) It considers the complete area under the membership function to determine the final crisp output, rather than selecting only the maximum membership value.
3) The crisp output is calculated by dividing the first moment of the membership function by its total area.

   Formula: $z* = \int z \times \mu(z)\, dz \,/\, \int \mu(z)\, dz$

   Here, z is the output variable, μ(z) is its membership value, and z* is the final crisp output.
4) The method provides a balanced result because it considers the shape and distribution of the entire membership function.
5) It is widely used in fuzzy control systems, such as temperature controllers, washing machines, and industrial automation.
6) Its main limitation is that calculating the centroid may require more computation than simpler methods, especially for complex membership functions.

Example: In a fuzzy temperature controller, the aggregated output represents different heating levels. The centroid method calculates the centre of this combined membership function to determine the final heating level.

2. Mean of Maximum (MOM) Method

1) The Mean of Maximum method identifies all output values that have the highest membership degree in the aggregated fuzzy

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1) The Mean of Maximum method identifies all output values that have the highest membership degree in the aggregated fuzzy output.
2) It calculates the arithmetic mean of these output values to obtain a single crisp output.
3) Unlike the centroid method, MOM considers only the values with maximum membership and ignores the remaining areas of the membership function.

   Formula: $z* = (z_1 + z_2 + \ldots + z_n) / n$

   Here, $z_1, z_2, \ldots, z_n$ are the output values with maximum membership, and n is the number of these values.
4) If the maximum membership occurs at only one output value, that value becomes the crisp output directly.
5) This method is simple to understand and computationally efficient because it requires only the maximum-membership values.
6) Its limitation is that it ignores the overall shape and area of the membership function, which may produce a less representative result in some situations.

Example: Suppose the maximum membership value occurs at output values 20, 30, and 40. The crisp output is calculated as:

$z* = (20 + 30 + 40) / 3$

$z* = 90 / 3 = 30$

Therefore, the final crisp output is 30.

Q.8) What are all the objects of using LSTEM? Explain its architecture.

Ans: Long Short-Term Memory (LSTM) is a special type of Recurrent Neural Network (RNN) designed to learn long-term dependencies in sequential data. It uses a memory cell and three gates to control the flow of information, helping overcome the vanishing gradient problem commonly encountered in traditional RNNs.

Objectives of Using LSTM

1) LSTM is designed to learn long-term dependencies by retaining important information from earlier time steps in a sequence.
2) It reduces the vanishing gradient problem by providing a controlled path for information and gradients through its cell state.
3) It selectively forgets irrelevant information using the forget gate, allowing the network to maintain useful context.
4) It controls the storage of new information through the input gate, helping the network update its memory when necessary.
5) It regulates the information passed to the next time step through the output gate, producing a suitable hidden state.
6) It is useful for sequential data in which the current output depends on previous inputs, such as text, speech, and time-series data.

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Architecture of LSTM

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The LSTM architecture consists of a cell state, a hidden state, and three gates: the forget gate, input gate, and output gate.

Architecture Diagram:

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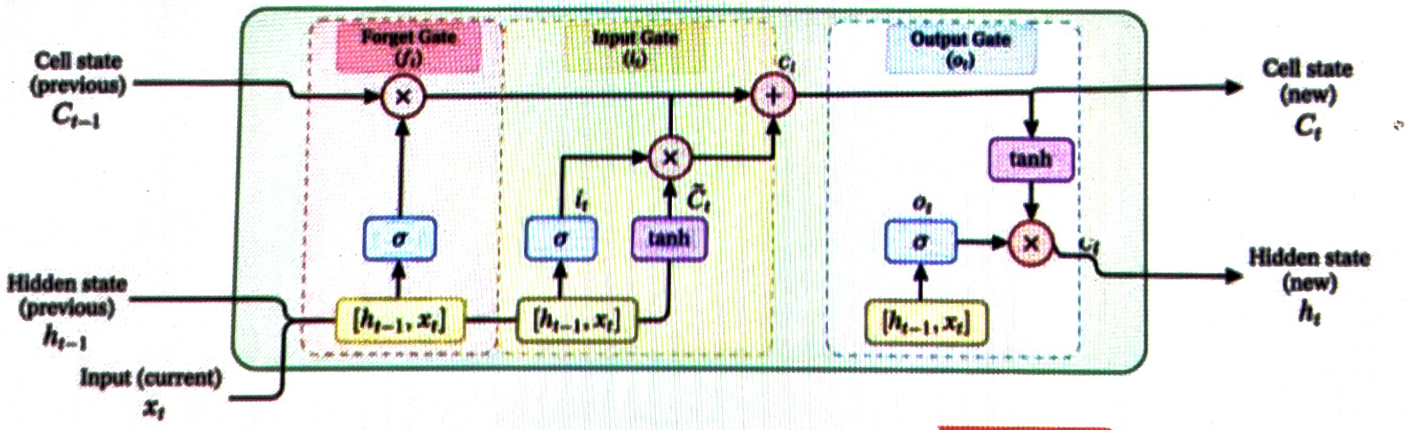

Main Components of LSTM

1) Forget Gate

1) The forget gate decides which information from the previous cell state should be retained or discarded.
2) It takes the current input (x_t) and previous hidden state (h_{t-1}) as inputs.

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2) It takes the current input (x_t) and previous hidden state (h_{t-1}) as inputs.

3) The sigmoid activation function produces values between 0 and 1, indicating how much information should be retained.

4) A value close to 0 means that information is mostly forgotten, while a value close to 1 means it is mostly retained.

5) This gate helps remove unnecessary information from memory while preserving useful context.

Formula: $f_t = \sigma(Wf \cdot [h_{t-1}, x_t] + bf)$

2) Input Gate

1) The input gate determines which new information should be stored in the cell state at the current time step.

2) It uses the current input and previous hidden state to decide which information is important.

3) The sigmoid function produces input-gate values between 0 and 1, controlling how much new information is added.

4) A tanh function generates candidate memory values between −1 and +1.

5) The input gate combines its values with the candidate memory to determine the new information added to the cell state.

Formulas: $i_t = \sigma(Wi \cdot [h_{t-1}, x_t] + bi), C_t = tanh(Wc \cdot [h_{t-1}, x_t] + bc)$

3) Cell State Update

1) The cell state carries important information across time steps and acts as the long-term memory of the LSTM.

2) The previous cell state is multiplied by the forget gate to remove information that is no longer useful.

3) The input gate controls how much candidate information is added to the existing memory.

4) This process allows the LSTM to preserve relevant information over long sequences and update it when new information arrives.

Formula: $C_t = f_t \times C_{t-1} + i_t \times C_t$. Here, × represents element-wise multiplication.

4) Output Gate

1) The output gate determines which information from the updated cell state should be passed to the hidden state.

2) It uses the current input and previous hidden state to calculate the output-gate values.

3) The sigmoid function controls the amount of information allowed to pass through the gate.

4) The updated cell state is passed through the tanh function, and its result is multiplied by the output-gate values.

5) The resulting hidden state is passed to the next time step and can also be used to generate the network's prediction.

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Formulas: $o_t = \sigma(Wo \cdot [h_{t-1}, x_t] + bo), h_t = o_t \times tanh(C_t)$

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Working of LSTM

1) The LSTM receives the current input (x_t), previous hidden state (h_{t-1}), and previous cell state (C_{t-1}).
2) The forget gate decides which information from the previous cell state should be retained or discarded.
3) The input gate selects useful new information and generates candidate values for updating the cell state.
4) The cell state is updated by combining the retained old information with the selected new information.
5) The output gate determines which information from the updated cell state forms the new hidden state.

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6) The new cell state and hidden state are passed to the next time step, allowing the network to learn patterns across a sequence.

Applications of LSTM

1) **Language Processing:** Used in language modelling, text generation, and machine translation to learn relationships between words across sequences.
2) **Speech Recognition:** Processes audio sequences to recognize spoken words and convert speech into text.
3) **Time-Series Forecasting:** Analyses historical data to predict values such as weather conditions, sales, and energy consumption.
4) **Anomaly Detection:** Identifies unusual patterns in sequential data, helping detect fraudulent transactions, network intrusions, and equipment failures.

Q.9) What is cognitive computing, and how does it differ from traditional computing? List and explain the process of building a cognitive application.

Ans: Cognitive computing is a branch of computing that uses technologies such as Artificial Intelligence (AI), Machine Learning (ML),

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Q.9) What is cognitive computing, and how does it differ from traditional computing? List and explain the process of building a cognitive application.

Ans: Cognitive computing is a branch of computing that uses technologies such as Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and data analytics to simulate human cognitive abilities. It enables systems to understand information, learn from experience, interpret unstructured data, and support decision-making.

Difference Between Cognitive Computing and Traditional Computing

Sr. No.	Cognitive Computing	Traditional Computing
1.	It attempts to simulate human thinking, learning, and reasoning.	It follows predefined instructions and programmed logic to process data.
2.	It can learn from data and improve its performance over time.	It generally requires explicit programming or updates to change its behaviour.
3.	It can process unstructured data such as text, images, and speech.	It is particularly effective with structured data and clearly defined inputs.
4.	It can interpret natural language and understand context to some extent.	It generally depends on predefined commands, rules, or input formats.
5.	It can handle uncertainty and provide recommendations based on available evidence.	It usually produces results according to predefined rules and algorithms.
6.	It supports human decision-making by presenting insights and possible solutions.	It mainly executes specified tasks and produces results based on programmed logic.

Process of Building a Cognitive Application

1) **Identify the Problem:** Define the problem the application must solve and identify the target users, objectives, and expected outcomes.
2) **Collect Data:** Gather relevant data from sources such as databases, documents, websites, images, and user interactions. The quality and relevance of data affect the application's performance.
3) **Prepare and Process Data:** Clean the collected data by removing errors, handling missing values, and organizing it into a suitable format for analysis and learning.
4) **Select Cognitive Technologies:** Choose appropriate technologies such as NLP for language understanding, ML for learning patterns, and computer vision for image analysis according to the application's requirements.

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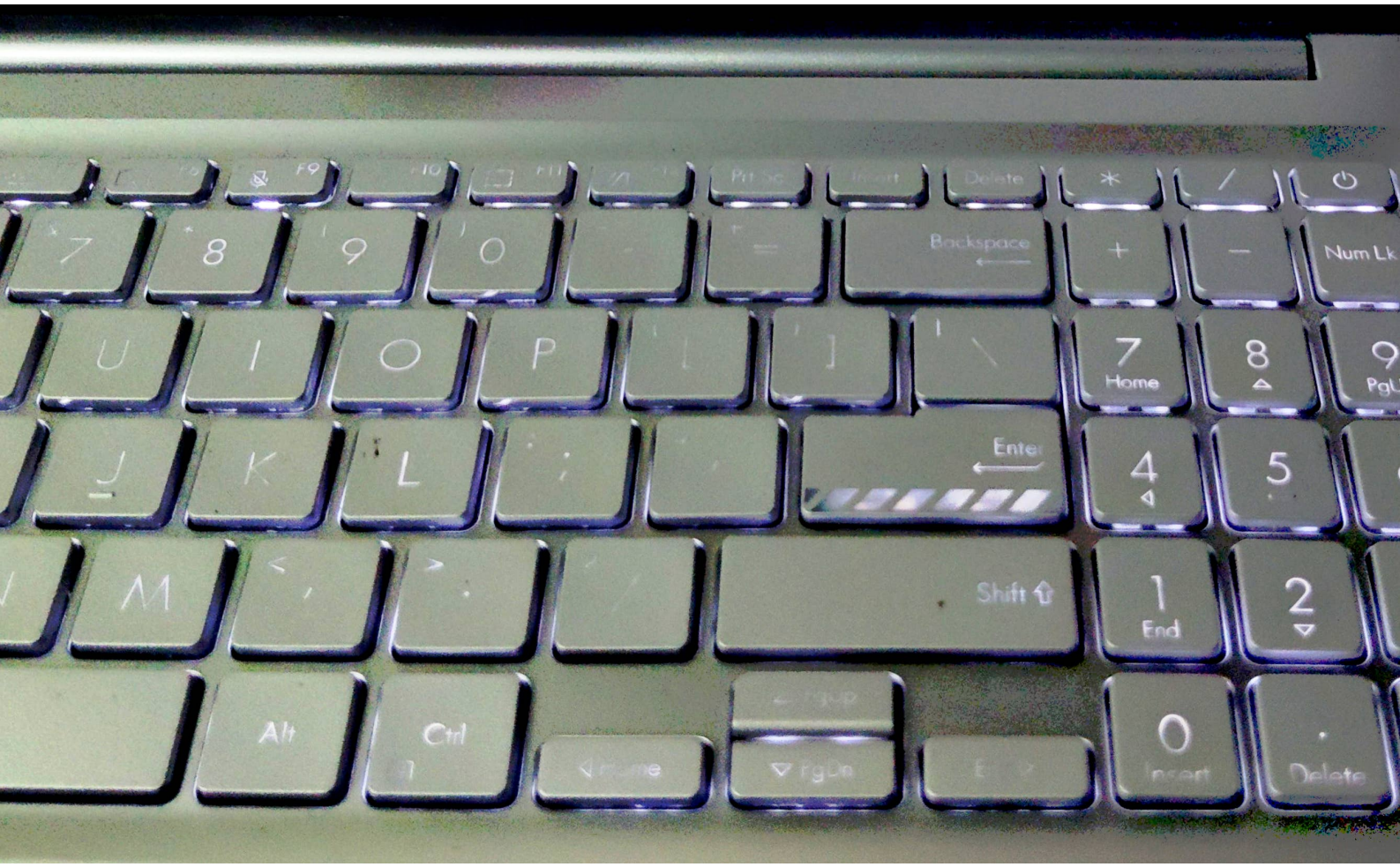
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3) **Prepare and Process Data:** Clean the collected data by removing errors, handling missing values, and organizing it into a suitable format for analysis and learning.

4) **Select Cognitive Technologies:** Choose appropriate technologies such as NLP for language understanding, ML for learning patterns, and computer vision for image analysis according to the application's requirements.

5) **Build the Knowledge Base:** Organize information using databases, taxonomies, or ontologies so that the application can identify relationships between concepts and retrieve relevant knowledge.

6) **Develop and Train Models:** Train suitable machine learning models using prepared data so that the application can recognize patterns, understand queries, and generate useful predictions or recommendations.

7) **Integrate Cognitive Services:** Connect the trained models with other application components, databases, APIs, and user interfaces to create a complete working system.

8) **Test and Evaluate:** Test the application using relevant examples and real-world scenarios. Evaluate its accuracy, reliability, response quality, and ability to handle different user queries.

9) **Deploy the Application:** Make the application available to users through a website, mobile application, cloud platform, or enterprise system.

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10) **Monitor and Improve:** Monitor application performance and user feedback, update the knowledge base, and retrain models when necessary to improve results over time.

Example: A cognitive customer-support application can understand questions written in natural language, search a knowledge base, learn from previous interactions, and recommend suitable solutions to users.

Q.10) Define Cognitive computing. Draw a neat diagram of components of the cognitive system and explain the components also List and explain the design principle of Cognitive system.

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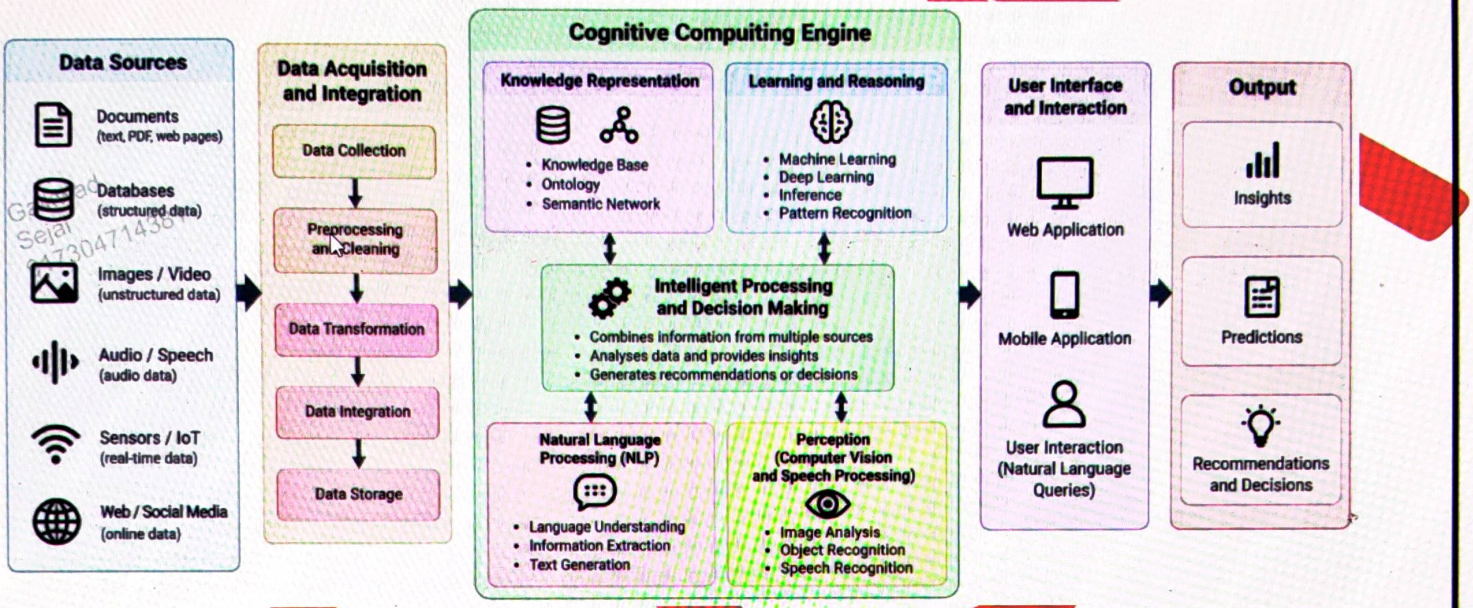
Q.10) Define Cognitive computing. Draw a neat diagram of components of the cognitive system and explain the components also List and explain the design principle of Cognitive system.

Ans: Cognitive computing is a branch of computing that uses technologies such as Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and data analytics to simulate human cognitive abilities. It enables systems to understand information, learn from experience, reason, and assist humans in solving complex problems and making decisions.

Components of a Cognitive System



1) **Data Sources:** These provide the information required by the cognitive system. Sources include documents, databases, images, videos, audio recordings, sensors, and web pages.
2) **Data Acquisition and Integration:** This component collects data from different sources, cleans it, removes errors, converts it into suitable formats, and combines it for further processing.
3) **Knowledge Representation:** It organizes information using knowledge bases, taxonomies, ontologies, and semantic networks. This helps the system identify relationships between concepts and retrieve relevant information.
4) **Learning and Reasoning:** Machine learning and deep learning identify patterns in data, while reasoning and inference help the system draw conclusions and generate useful insights.
5) **Natural Language Processing (NLP):** NLP enables the system to understand human language, extract meaning from text,

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5) **Natural Language Processing (NLP):** NLP enables the system to understand human language, extract meaning from text, interpret queries, and generate meaningful responses.

6) **Perception:** This component processes visual and audio information using computer vision and speech-processing techniques. It can recognize objects, images, patterns, and spoken words.

7) **Intelligent Processing and Decision-Making:** It combines information from different components, analyses possible outcomes, and generates insights, recommendations, or decisions to assist users.

8) **User Interface and Interaction:** It provides a way for users to communicate with the cognitive system through websites, mobile applications, chat interfaces, or voice-based interactions.

9) **Output:** This component presents the results generated by the system, such as predictions, insights, recommendations, answers, and decision support.

Design Principles of a Cognitive System

1) **Human-Centric Design:** The system should support and enhance human capabilities. It should provide useful information and recommendations while allowing humans to make important decisions when required.

2) **Learning from Data:** The system should learn from data, previous interactions, and feedback to improve its performance and provide increasingly relevant results.

3) **Handling Uncertainty:** It should handle incomplete, ambiguous, or noisy information and provide reasonable conclusions while considering uncertainty in the available data.

4) **Context Awareness:** The system should understand the context of a query, including relevant information and user requirements, to provide appropriate responses.

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5) **Scalability:** The system should be capable of handling increasing amounts of data, growing numbers of users, and additional computational requirements without significant performance degradation.

6) **Adaptability:** It should adapt to changing data, user behaviour, and environmental conditions without requiring complete

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6) **Adaptability:** It should adapt to changing data, user behaviour, and environmental conditions without requiring complete redevelopment whenever circumstances change.

7) **Transparency and Explainability:** The system should provide understandable explanations for its predictions and recommendations so users can evaluate and trust its results.

8) **Interdisciplinary Integration:** It should combine technologies such as AI, machine learning, NLP, computer vision, and data analytics to solve complex problems effectively.

9) **Continuous Improvement:** The system should be monitored regularly and improved using new data, user feedback, performance evaluation, and model updates.

10) **Ethical and Responsible Use:** The system should protect user privacy, maintain data security, reduce bias, and ensure that its decisions are fair and responsible.

Q.11) Briefly explain Data science for Multimodal applications.

Ans: Data science for multimodal applications involves collecting, processing, and analysing data from multiple sources or modalities, such as text, images, video, and audio. It combines statistical techniques, machine learning, and deep learning to extract meaningful information and understand complex situations more effectively than analysing a single data type alone.

1) **Text Data:** Text analytics and Natural Language Processing (NLP) are used to analyse documents, emails, reviews, and social media posts. Applications include sentiment analysis, text classification, language translation, and information extraction.
2) **Image Data:** Image data is analysed using image processing and computer vision techniques to identify objects, patterns, and visual features. Applications include facial recognition, medical image analysis, and object detection.
3) **Video Data:** Video analytics processes sequences of image frames to identify objects, activities, movements, and events over time. It is commonly used in traffic monitoring, security surveillance, and activity recognition.
4) **Audio Data:** Audio analytics processes speech, music, and environmental sounds to identify spoken words, recognize speakers, and detect sound patterns. Applications include speech recognition, voice assistants, and audio classification.
5) **Multimodal Data Integration:** This process combines information from two or more modalities, such as text with images or audio with video. The combined information provides a more complete understanding of a situation and can improve prediction and decision-making.
6) **Feature Extraction and Analysis:** Relevant features are extracted from each data type using suitable techniques. Machine learning models then analyse these features to identify relationships, patterns, and useful insights across different modalities.
7) **Model Training and Evaluation:** Machine learning or deep learning models are trained using multimodal datasets. Their performance is evaluated using suitable metrics to check accuracy, reliability, and effectiveness for the intended application.

Applications of Multimodal Data Science

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Applications of Multimodal Data Science

1) **Healthcare:** Combines medical reports, X-rays, and patient records to support disease diagnosis.
2) **Autonomous Vehicles:** Uses camera images, video, and sensor data to detect obstacles, pedestrians, and traffic signs.
3) **Virtual Assistants:** Combines speech, text, and images to understand user requests and provide relevant responses.
4) **Security and Surveillance:** Analyses video footage and audio signals to detect suspicious activities.
5) **Recommendation Systems:** Uses images, text reviews, and user preferences to recommend relevant products and content.
6) **Education:** Combines text, audio, and video to develop interactive learning systems and personalized educational content.
7) **E-commerce:** Analyses product images, descriptions, and customer reviews to improve product search and recommendations.
8) **Emotion Recognition:** Combines facial expressions, voice, and text to identify possible emotional states.
9) **Social Media Analysis:** Analyses posts, images, videos, and comments to understand trends and user behaviour.
10) **Disaster Management:** Combines satellite images, videos, sensor readings, and reports to assess damage and support emergency response.

Q.12) How bagging and boosting handle bias-variance trade-off differently, and analyze their effectiveness in dealing with noisy data and overfitting. Explain with algorithmic such as Random forest and AdaBoost.

Ans: Bagging and boosting are ensemble learning techniques that combine multiple models to improve prediction accuracy. Bagging mainly reduces variance, whereas boosting primarily reduces bias by learning from the errors of previous models.

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1. Bagging (Bootstrap Aggregating)

1) Bagging creates multiple training datasets by randomly sampling the original dataset with replacement. A separate model is trained on each sample.
2) The models are trained independently, and their predictions are combined using majority voting for classification or averaging for regression.
3) Bagging mainly reduces variance by combining predictions from multiple models, making the final result less sensitive to changes in training data.
4) It is generally more resistant to noise because the influence of individual noisy observations can be reduced through aggregation.
5) Bagging helps control overfitting, especially when high-variance models such as decision trees are used as base learners.

2. Boosting

1) Boosting trains models sequentially, where each new model focuses on correcting errors made by previous models.
2) It combines multiple weak learners using weighted voting or a weighted sum to produce a stronger predictive model.
3) Boosting primarily reduces bias by gradually improving the model's ability to learn patterns that earlier models missed.
4) It can be sensitive to noisy data because misclassified observations may receive greater importance during training.

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5) Excessive boosting rounds or insufficient regularization may cause overfitting, particularly when the dataset contains mislabeled observations or outliers.

3. Random Forest Algorithm (Bagging)

Random Forest is an ensemble learning algorithm that combines multiple decision trees. It uses bootstrap sampling and random feature selection to reduce the correlation between trees and improve prediction accuracy.

Algorithm:

1) Draw multiple bootstrap samples from the original training dataset.
2) Train a separate decision tree on each bootstrap sample.
3) At each split, select the best split from a randomly selected subset of features.
4) Repeat the process to construct multiple decision trees independently.
5) Combine the predictions using majority voting for classification or averaging for regression.

Effectiveness: Random Forest mainly reduces variance and is generally less prone to overfitting than a single decision tree. It can handle noisy data relatively well, although severe noise may still affect its predictions.

4. AdaBoost Algorithm (Boosting)

AdaBoost (Adaptive Boosting) combines multiple weak learners, commonly shallow decision trees called decision stumps. It assigns greater importance to misclassified observations so that subsequent learners focus on difficult examples.

Algorithm:

1) Assign equal weights to all training observations initially.
2) Train a weak learner using the weighted training data.
3) Calculate the weighted classification error of the learner.
4) Assign greater influence to more accurate learners when combining their predictions.
5) Increase the relative weights of misclassified observations and decrease the relative weights of correctly classified observations.
6) Train the next weak learner using the updated observation weights.
7) Repeat the process and combine the learners using weighted voting to obtain the final prediction.

Effectiveness: AdaBoost primarily reduces bias by progressively correcting classification errors. However, it can be sensitive to noisy data because mislabeled observations may repeatedly receive higher weights, increasing the risk of overfitting.

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Effectiveness: AdaBoost primarily reduces bias by progressively correcting classification errors. However, it can be sensitive to noisy data because mislabeled observations may repeatedly receive higher weights, increasing the risk of overfitting.

5. Handling the Bias-Variance Trade-off, Noise, and Overfitting

1) **Bagging:** It reduces variance by averaging predictions from independently trained models, making it suitable for high-variance learners.
2) **Boosting:** It primarily reduces bias by focusing on errors made by previous learners, helping improve models that initially underfit the data.

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3) **Noisy Data:** Bagging is generally more robust to noise, while boosting may give excessive attention to mislabeled observations and outliers.
4) **Overfitting:** Bagging controls overfitting through aggregation, whereas boosting may require control of the number of learners, learning rate, and model complexity to avoid overfitting.

Q.13) Explain ensemble methods in details.

Ans: Ensemble methods are machine learning techniques that combine predictions from multiple models to produce more accurate and reliable results than a single model. They improve performance by combining the strengths of different learners and reducing prediction errors.

Types of Ensemble Methods

1. Bagging (Bootstrap Aggregating)

1) Bagging creates multiple training datasets by randomly sampling the original dataset with replacement. Each dataset is used to train a separate model independently.
2) The models are trained in parallel, and their predictions are combined using majority voting for classification or averaging for regression.

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1. Bagging (Bootstrap Aggregating)

1) Bagging creates multiple training datasets by randomly sampling the original dataset with replacement. Each dataset is used to train a separate model independently.
2) The models are trained in parallel, and their predictions are combined using majority voting for classification or averaging for regression.
3) Bagging mainly reduces variance by combining predictions from multiple models, making the final result less sensitive to variations in training data.
4) It helps control overfitting, particularly when high-variance models such as decision trees are used as base learners.
5) It is generally more robust to noise because the effect of individual prediction errors can be reduced through aggregation.
6) Bagging works best when the individual models are diverse and their prediction errors are not highly correlated.

Example: Random Forest trains multiple decision trees on bootstrap samples and uses random feature selection to improve prediction performance.

2. Boosting

1) Boosting trains models sequentially, where each new model focuses on correcting errors made by previous models.
2) It combines multiple weak learners using weighted voting or a weighted sum to produce the final prediction.
3) Boosting primarily reduces bias by gradually improving weak learners and capturing patterns that earlier models missed.
4) In algorithms such as AdaBoost, misclassified observations receive higher relative weights so that subsequent learners focus more on difficult examples.
5) Boosting can achieve high prediction accuracy by combining several simple models into a stronger ensemble.
6) It may be sensitive to noisy data and outliers because incorrectly labelled observations can repeatedly receive greater importance during training.
7) Excessive boosting rounds or insufficient regularization can cause overfitting, so the number of learners and model complexity should be controlled.

Example: AdaBoost trains weak learners sequentially, updates observation weights after each round, and combines the learners using weighted voting.

3. Stacking (Stacked Generalization)

1) Stacking combines predictions from different types of base models using another model called a meta-learner.

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1) Stacking combines predictions from different types of base models using another model called a meta-learner.
2) Base models may include decision trees, support vector machines, and logistic regression, allowing different algorithms to learn different patterns in the data.
3) The meta-learner learns how to combine the base models' predictions to generate the final result.
4) Properly generated out-of-fold predictions are commonly used to train the meta-learner and reduce data leakage and overfitting.
5) Stacking can improve prediction accuracy when the base models make different types of errors and provide complementary information.
6) It can be used for both classification and regression by selecting an appropriate meta-learner for the problem.

Example: Predictions from a decision tree and an SVM can be combined using logistic regression as the meta-learner.

4. Voting

1) Voting combines predictions from multiple classification models to determine the final class.
2) In hard voting, each model votes for a class, and the class receiving the most votes is selected.

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3) In soft voting, the class probabilities predicted by the models are averaged, and the class with the highest average probability is selected.
4) Soft voting generally works best when the models provide reasonably calibrated class probabilities.
5) Voting is simple to implement because it does not require training an additional meta-learner.
6) It often performs well when the individual models have reasonably good accuracy and make different prediction errors.
7) The contribution of individual models can be adjusted using weighted voting, where more reliable models receive greater influence on the final prediction.

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7) The contribution of individual models can be adjusted using weighted voting, where more reliable models receive greater influence on the final prediction.

Example: Logistic regression, a decision tree, and an SVM can combine their predictions to classify emails as spam or not spam.

Applications of Ensemble Methods

1) **Healthcare:** Predicting diseases using patient records, clinical measurements, and medical data.
2) **Finance:** Detecting fraudulent transactions, assessing credit risk, and predicting financial outcomes.
3) **Image Classification:** Improving object recognition and image classification accuracy.
4) **Forecasting:** Predicting sales, customer demand, and other business outcomes.
5) **Cybersecurity:** Detecting suspicious activities, identifying malicious behaviour, and classifying potential threats.

Q.14) Perform a case study on video recommendation systems. (data science based)

Ans:

1. Introduction: A video recommendation system is a data science application that suggests videos to users based on their interests, viewing history, ratings, searches, and behaviour. Platforms such as YouTube and Netflix use recommendation systems to provide personalized content and improve user engagement.

2. Objectives of the Case Study

1) To understand how data science is used to recommend videos to users.
2) To analyze user behaviour and identify viewing preferences.
3) To provide personalized video recommendations.
4) To improve user engagement, satisfaction, and content discovery.
5) To increase watch time and help users find relevant videos efficiently.

3. Data Collection

The system collects and analyzes different types of data, such as:

1) **User Data:** User ID, age group, language preferences, and selected interests, where available.
2) **Viewing History:** Previously watched videos, watch duration, and completed or skipped videos.
3) **Interaction Data:** Likes, dislikes, ratings, comments, shares, and subscriptions.

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3) **Interaction Data:** Likes, dislikes, ratings, comments, shares, and subscriptions.
4) **Video Data:** Video title, category, description, tags, language, and upload date.
5) **Search Data:** Keywords and topics searched by the user.

4. Data Preprocessing

1) The collected data is cleaned by removing duplicate records and handling missing or incorrect values.
2) User interactions are organized into a structured format for analysis.
3) Video titles, descriptions, and tags are processed to identify important keywords and topics.
4) User activity is converted into useful features, such as watch time, preferred categories, and frequency of interaction.

5. Data Science Techniques Used

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1) **Collaborative Filtering:** Recommends videos based on the preferences of users with similar viewing behaviour. For example, if users with similar interests watch a particular video, the system may recommend it to another user with similar preferences.
2) **Content-Based Filtering:** Recommends videos similar to those the user has watched or liked. It uses features such as video category, title, tags, and description.
3) **Machine Learning:** Algorithms learn patterns from user interactions and predict which videos a user is likely to watch or enjoy.

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4) **Natural Language Processing (NLP):** Analyzes video titles, descriptions, captions, and tags to understand the topics covered in videos.
5) **Ranking Algorithms:** Assign scores to candidate videos based on predicted relevance and other platform objectives, then arrange them in a recommended order.

6. Working of the Video Recommendation System

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5) **Ranking Algorithms:** Assign scores to candidate videos based on predicted relevance and other platform objectives, then arrange them in a recommended order.

6. Working of the Video Recommendation System

1) The system collects user activity and video-related data.

2) The collected data is cleaned and processed to extract useful features.

3) The system identifies user interests using viewing history, interactions, and content preferences.

4) Recommendation algorithms generate a list of potentially relevant videos.

5) A ranking model orders the videos according to their predicted relevance to the user.

6) The recommended videos are displayed on the user's home page or in the suggested-video section.

7) The system collects new feedback from the user's interactions and updates future recommendations.

7. Example Case Study: YouTube

YouTube is an example of a video platform that uses machine learning and large-scale data analysis to personalize recommendations.

1) When a user watches educational videos on Python programming, the system can identify programming as a possible interest.

2) It may recommend related videos on Python projects, data structures, or programming tutorials.

3) If the user frequently watches videos from a particular category, that behaviour can influence future recommendations.

4) The system can use signals such as watch history, clicks, watch duration, and feedback to estimate which videos may be relevant.

5) Recommendations may change as the user's interests and viewing behaviour change.

8. Evaluation of the System

1) **Click-Through Rate (CTR):** Measures the proportion of displayed recommendations that users click.

$$CTR = (Number\ of\ clicks\ /\ Number\ of\ impressions) \times 100$$

2) **Watch Time:** Measures how much time users spend watching recommended videos.

3) **Engagement Rate:** Measures interactions such as likes, shares, comments, and subscriptions.

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3) **Engagement Rate:** Measures interactions such as likes, shares, comments, and subscriptions.

4) **Recommendation Relevance:** Evaluates how closely recommended videos match users' interests, using suitable offline metrics or user feedback.

5) **User Satisfaction:** Evaluates whether recommendations help users discover useful or enjoyable content.

9. Challenges

1) **Cold-Start Problem:** New users or newly uploaded videos may have insufficient interaction data for accurate recommendations.

2) **Data Sparsity:** Many users interact with only a small fraction of available videos, making preference prediction difficult.

3) **Privacy:** User activity must be handled responsibly and according to applicable privacy requirements.

4) **Filter Bubbles:** Recommending similar content repeatedly may limit exposure to different topics and viewpoints.

5) **Changing Interests:** A user's preferences may change over time, so recommendations need regular updating.

10. Conclusion: Video recommendation systems demonstrate how data science combines data preprocessing, machine learning, user behaviour analysis, and ranking algorithms to deliver personalized content. These systems improve content discovery and user experience, but they must also address privacy, recommendation quality, and content diversity.

Q.15) Describe Deep Learning concept with an example. State and elaborate the applications of deep learning.

Ans:

1) Deep Learning is a subfield of Machine Learning that uses artificial neural networks with multiple layers to learn complex patterns from large amounts of data. It automatically extracts important features from data such as images, text, audio, and video, reducing the need for manual feature extraction.
2) Deep learning is inspired by the structure and functioning of the human brain. It uses interconnected artificial neurons arranged in input, hidden, and output layers to process information and generate predictions.

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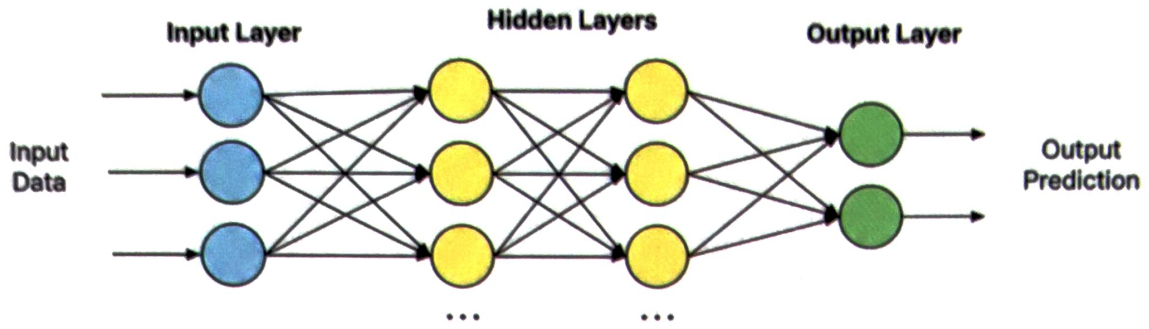
Architecture of Deep Learning



1) **Input Layer:** Receives input data such as images, text, or numerical values.
2) **Hidden Layers:** Process the input through multiple layers and learn simple to complex patterns using weights, biases, and activation functions.
3) **Output Layer:** Produces the final result, such as a predicted class, numerical value, or generated content.
4) **Training Process:** The network compares its prediction with the actual result, calculates the error using a loss function, and updates its weights using backpropagation and an optimization algorithm.

Example of Deep Learning: Image Recognition

Consider a deep learning model trained to identify cats and dogs from images.

1) The input layer receives an image containing a cat or a dog.
2) The initial hidden layers learn basic features such as edges, lines, and colours.
3) The deeper layers identify complex features such as ears, eyes, facial shapes, and body structures.
4) The output layer predicts whether the image contains a cat or a dog.
5) During training, the model learns from labelled images and adjusts its weights to reduce prediction errors.

Thus, after sufficient training, the model can classify new images that it has not seen before.

Applications of Deep Learning

1) **Computer Vision:** Deep learning enables computers to understand and analyze images and videos. Convolutional Neural Networks (CNNs) are commonly used for image classification, object detection, and facial recognition.
Example: Detecting vehicles and pedestrians in self-driving cars.

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Example: Detecting vehicles and pedestrians in self-driving cars.

2) **Natural Language Processing (NLP):** Deep learning helps computers understand, process, and generate human language. It is used in language translation, text summarization, sentiment analysis, and conversational systems.
   Example: Translating a sentence from English to Hindi using a neural machine translation system.
3) **Speech Recognition:** Deep learning converts spoken language into text by learning patterns from audio signals. It can also help identify spoken words in different accents and speaking conditions.
   Example: Voice typing and speech-to-text systems.
4) **Healthcare and Medical Diagnosis:** Deep learning analyzes medical images and patient data to help healthcare professionals identify possible diseases. It can support the detection of abnormalities in X-rays, CT scans, and MRI images.
   Example: Detecting possible tumours in medical images to assist doctors in diagnosis.
5) **Autonomous Vehicles:** Deep learning processes data from cameras and other sensors to recognize roads, traffic signs, pedestrians, and nearby vehicles. This information supports decisions related to navigation and driving.
   Example: A self-driving car identifying a pedestrian crossing the road.
6) **Recommendation Systems:** Deep learning analyzes user behaviour, preferences, and interactions to recommend relevant content or products. It can learn complex relationships between users and available items.
   Example: YouTube recommending videos based on a user's viewing history and interests.
7) **Fraud Detection and Banking:** Deep learning identifies unusual patterns in financial transactions and can help detect potentially fraudulent activities. It learns from historical transaction data and identifies suspicious behaviour.
   Example: A banking system flagging an unusual transaction for further verification.
8) **Generative AI:** Deep learning models can generate new content, including text, images, music, audio, and video. These models learn patterns from training data and use them to create new outputs.
   Example: AI tools generating images from text descriptions or producing text-based answers.
9) **Robotics:** Deep learning helps robots recognize objects, understand their surroundings, and perform tasks based on visual or sensor information. It can also support object manipulation and navigation.
   Example: A warehouse robot identifying and picking products from shelves.
10) **Weather Forecasting:** Deep learning models can analyze historical weather observations and other environmental data to identify patterns and support weather prediction. Their performance depends on the quality of data and the forecasting task.
    Example: Predicting rainfall or estimating the likelihood of severe weather.

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11) **Cybersecurity:** Deep learning can analyze network traffic, system logs, and user activity to identify suspicious patterns. It helps security systems detect possible malware, intrusions, and abnormal behaviour.
Example: Detecting unusual network activity that may indicate a cyberattack.

Q.16) Explain various metrics for evaluating the classifier performance. Discuss any two cross validation methods.

Ans: Classifier evaluation metrics are used to measure how effectively a classification model predicts the correct class labels. They help determine the accuracy, reliability, and effectiveness of a model and identify the types of errors it makes. The various metrics for evaluating classifier performance are as follows:

1) **Accuracy:** Accuracy measures the proportion of total predictions that are correctly classified by the model. It is most useful when the dataset has a reasonably balanced distribution of classes.
Formula: $Accuracy = (TP + TN) / (TP + TN + FP + FN)$
2) **Precision:** Precision measures the proportion of instances predicted as positive that are actually positive. High precision indicates that the model produces fewer false-positive predictions.
Formula: $Precision = TP / (TP + FP)$
3) **Recall (Sensitivity):** Recall measures the proportion of actual positive instances that are correctly identified by the classifier. It is important when missing a positive case can have serious consequences, such as disease detection.
Formula: $Recall = TP / (TP + FN)$
4) **F1-Score:** F1-score is the harmonic mean of precision and recall. It is useful when both false-positive and false-negative errors need to be considered, especially for imbalanced datasets.
Formula: $F1 - Score = 2 \times (Precision \times Recall) / (Precision + Recall)$
5) **Specificity:** Specificity measures the proportion of actual negative instances that are correctly classified as negative. It indicates how effectively the model avoids incorrectly classifying negative cases as positive.
Formula: $Specificity = TN / (TN + FP)$
6) **Confusion Matrix:** A confusion matrix is a table that compares the actual class labels with the predicted class labels. For binary classification, it contains four values:

- **True Positive (TP):** The model correctly predicts the positive class.
- **True Negative (TN):** The model correctly predicts the negative class.
- **False Positive (FP):** The model predicts positive when the actual class is negative.
- **False Negative (FN):** The model predicts negative when the actual class is positive.

7) **ROC Curve and AUC:** The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate against the False Positive Rate at different classification thresholds. The Area Under the Curve (AUC) summarizes how well the classifier distinguishes between positive and negative classes; a higher AUC generally indicates better discrimination.
$TPR = TP / (TP + FN)$
$FPR = FP / (FP + TN)$

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Cross-Validation Methods

Cross-validation is a model evaluation technique in which a dataset is divided into training and validation subsets multiple times. It helps estimate how well a machine learning model will perform on unseen data and reduces dependence on a single train-test split. Two cross-validation methods are explained below.

1. **K-Fold Cross-Validation:** K-Fold Cross-Validation divides the dataset into K approximately equal-sized folds. The model is trained and evaluated K times, with each fold used once for validation while the remaining folds are used for training.

   Steps:

   1) Divide the dataset into K approximately equal-sized folds.
   2) Select one fold as the validation set and use the remaining K − 1 folds to train the model.
   3) Evaluate the trained model using the selected validation fold.
   4) Repeat the process K times, changing the validation fold in every iteration.
   5) Calculate the average of the K evaluation scores to estimate the model's performance.

   Example: Suppose a dataset contains 1,000 records and K = 5. The data is divided into five folds of 200 records each. In each iteration, 800 records are used for training and 200 for validation. After five iterations, the average score is calculated.

   Advantages:

   - Every record is used for validation once and for training in the other iterations.
   - It provides a more reliable performance estimate than a single train-validation split in many situations.

2. **Stratified K-Fold Cross-Validation:** Stratified K-Fold Cross-Validation is a variation of K-Fold Cross-Validation that maintains approximately the same proportion of each class in every fold as in the original dataset. It is particularly useful for classification problems involving imbalanced datasets.

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Steps:

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Steps:

1) Identify the class labels and determine the proportion of each class in the dataset.
2) Divide the dataset into K folds while maintaining approximately the same class proportions in each fold.
3) Select one fold for validation and use the remaining K – 1 folds for training.
4) Repeat the process until every fold has been used once for validation.
5) Calculate the average evaluation score across all K iterations.

Example: Suppose a dataset contains 1,000 records, of which 800 belong to the negative class and 200 belong to the positive class. In 5-fold stratified cross-validation, each fold will contain approximately 160 negative records and 40 positive records. This preserves the 80:20 class distribution in each fold.

Advantages:

- It ensures that each fold contains a representative proportion of the classes, where feasible.
- It is particularly suitable for imbalanced classification problems because a rare class is less likely to be missing from a validation fold.

Q.17) What is activation function? Discuss different types of activation function in Neural networks.

Ans: Activation Function

1) An activation function is a mathematical function used in an artificial neural network to determine the output of a neuron based on its weighted inputs and bias.
2) It introduces non-linearity into the network, allowing it to learn complex patterns and relationships in data.
3) Without non-linear activation functions, a neural network consisting of multiple layers would be limited in its ability to learn complex relationships.

Types of Activation Functions

1) **Binary Step Function:** The binary step function produces either 0 or 1 depending on whether the input crosses a specified threshold. It is simple but is not commonly used in modern neural networks because it does not provide a useful gradient for gradient-based learning.
   Formula: $f(x) = 1, if\ x \geq 0;\ f(x) = 0, if\ x < 0$.
2) **Linear Function:** The linear activation function produces an output directly proportional to its input. It is mainly used in the output layer of regression models, where the predicted value can be continuous.
   Formula: $f(x) = x$
2) **Sigmoid Function:** The sigmoid function converts any real-valued input into an output between 0 and 1. It is commonly used

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3) **Sigmoid Function:** The sigmoid function converts any real-valued input into an output between 0 and 1. It is commonly used in the output layer of binary classification models to represent the predicted probability of the positive class.
   Formula: $f(x) = 1 / (1 + e^{-x})$
4) **Tanh Function:** The hyperbolic tangent function converts the input into a value between −1 and +1. Its output is centred around zero, but it may suffer from the vanishing gradient problem for very large positive or negative inputs.
   Formula: $f(x) = (e^{x} - e^{-x}) / (e^{x} + e^{-x})$
5) **ReLU (Rectified Linear Unit):** ReLU produces zero for negative inputs and returns the input itself for positive inputs. It is widely used in hidden layers because it is computationally simple and helps reduce the vanishing gradient problem in many situations.
   Formula: $f(x) = max(0, x)$
6) **Leaky ReLU:** Leaky ReLU is a variation of ReLU that allows a small, non-zero output for negative inputs. It helps reduce the problem of neurons becoming inactive when their inputs remain negative.
   Formula: $f(x) = x, if\ x \geq 0;\ f(x) = \alpha x, if\ x < 0$
   Here, α is a small positive constant, commonly 0.01.
7) **Softmax Function:** The softmax function converts a set of numerical values into probabilities whose sum equals 1. It is commonly used in the output layer of multiclass classification networks to estimate the probability of each class.
   Formula: $Softmax(z_i) = e^{zi} - \Sigma_j\, e^{zj}$
   Here, z_i is the input score for class i, and the denominator is the sum of exponentials of the scores for all classes.
8) **ELU (Exponential Linear Unit):** ELU returns the input itself for positive values and produces a smooth negative output for negative values. It can help maintain negative activations and may improve learning in some neural network architectures.
   Formula: $f(x) = x, if\ x \geq 0;\ f(x) = \alpha(e^{x} - 1), if\ x < 0$
   Here, α is a positive constant.
9) **Swish Function:** Swish is a smooth, non-linear activation function that multiplies the input by its sigmoid value. It is used in some deep learning architectures and can perform well in certain tasks.
   Formula: $f(x) = x \times sigmoid(x)$
   where $\text{sigmoid}(x) = 1 / (1 + e^{-x})$

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10) **10. GELU (Gaussian Error Linear Unit):** GELU smoothly scales the input according to its value and is used in many modern deep learning models, particularly Transformer-based architectures. Unlike ReLU, it does not sharply set all negative inputs to zero.

Approximate formula: $f(x) \approx 0.5x[1 + tanh(\sqrt{(2/\pi)}(x + 0.044715x^3))]$

Q.18) List three industries where data science is extensively used and briefly describe one application in each industry.

Ans: Data science is used in various industries to analyze large amounts of data, identify patterns, make predictions, and support decision-making. It helps organizations improve efficiency, reduce costs, manage risks, and provide better services. Three major industries where data science is extensively used are explained below.

1. Healthcare Industry

The healthcare industry uses data science to analyze patient records, medical images, laboratory reports, and clinical data. It helps doctors identify diseases, predict health risks, and make informed treatment decisions.

1) **Disease Prediction:** Machine learning models analyze patient history, symptoms, and medical records to predict the risk of diseases such as diabetes and heart disease.
2) **Medical Image Analysis:** Data science techniques help analyze X-rays, CT scans, and MRI images to identify abnormalities and assist doctors in diagnosis.
3) **Personalized Treatment:** Patient data can be analyzed to help healthcare professionals select suitable treatments according to individual patient characteristics.
4) **Drug Discovery:** Data science helps researchers analyze biological and chemical data to identify promising drug candidates and accelerate research.
5) **Patient Monitoring:** Data collected from wearable devices and medical sensors can be analyzed to monitor vital signs and identify unusual changes.
6) **Hospital Management:** Hospitals use data analysis to predict patient admissions, manage beds, allocate staff, and improve the use of medical resources.

Application: Disease Prediction

A machine learning model analyzes a patient's age, blood glucose level, weight, and medical history to estimate the risk of diabetes. This information can help healthcare professionals identify high-risk patients and recommend further medical assessment.

2. Banking and Finance Industry

The banking and finance industry uses data science to analyze financial transactions, assess creditworthiness, detect fraud,

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The banking and finance industry uses data science to analyze financial transactions, assess creditworthiness, detect fraud, and support investment decisions. It helps financial institutions improve security and manage financial risks.

1) **Fraud Detection:** Machine learning models analyze transaction patterns to identify unusual activities that may indicate fraudulent transactions.
2) **Credit Risk Assessment:** Banks analyze customers' income, repayment history, and existing debts to estimate the risk of lending money.
3) **Customer Segmentation:** Data science groups customers according to their spending habits, financial needs, and transaction patterns so that banks can provide suitable services.
4) **Investment Analysis:** Financial analysts use historical market data and statistical models to study trends and support investment decisions.
5) **Algorithmic Trading:** Data-driven algorithms analyze market information and execute trades according to predefined strategies and conditions.
6) **Risk Management:** Data science models help financial institutions estimate potential losses and evaluate different financial risks.
7) **Customer Service:** Chatbots and virtual assistants use natural language processing to answer common banking queries and provide basic assistance.

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Application: Fraud Detection

A bank's fraud detection system analyzes transaction amount, location, frequency, and spending patterns. If a transaction differs significantly from a customer's usual behaviour, the system may flag it for verification or additional security checks.

3. E-commerce and Retail Industry

The e-commerce and retail industry uses data science to understand customer behaviour, analyze purchasing patterns, forecast demand, and improve the shopping experience. It helps businesses make better decisions about products, pricing, inventory, and marketing.

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1) **Recommendation Systems:** Data science analyzes browsing history, previous purchases, and customer preferences to recommend relevant products.
2) **Demand Forecasting:** Historical sales data and seasonal trends are analyzed to estimate future product demand.
3) **Customer Behaviour Analysis:** Businesses study customers' searches, clicks, purchases, and shopping patterns to understand their needs.
4) **Dynamic Pricing:** Data-driven systems can adjust product prices according to demand, competition, inventory, and other business factors.
5) **Inventory Management:** Data science helps retailers maintain suitable stock levels and reduce overstocking or product shortages.
6) **Customer Segmentation:** Customers are grouped according to purchasing behaviour, preferences, and spending patterns to support targeted marketing.
7) **Sentiment Analysis:** Natural language processing analyzes customer reviews and feedback to identify opinions about products and services.

Application: Recommendation Systems

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An online shopping platform analyzes a customer's browsing history, previous purchases, and product preferences to recommend relevant products. For example, if a customer frequently searches for laptops, the platform may recommend laptops, laptop bags, and related accessories.

Thus, data science enables healthcare, banking and finance, and e-commerce industries to make data-driven decisions, improve efficiency, reduce risks, and deliver better services.

Q.19) Discuss the limitation of deterministic approaches and how probability reasoning address these limitation with example.

Ans: Deterministic Approach and Probability Reasoning

1) A deterministic approach is a method in which the same input always produces the same output under the same conditions. It assumes that the information and rules required to solve a problem are known and certain.
2) In real-world situations, information may be incomplete, uncertain, or affected by unpredictable events. Probability reasoning addresses these limitations by representing uncertainty using probabilities and estimating the likelihood of different outcomes.

Limitations of Deterministic Approaches

1) **Inability to Handle Uncertainty:** Deterministic approaches assume that the input information is certain and cannot

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Limitations of Deterministic Approaches

1) **Inability to Handle Uncertainty:** Deterministic approaches assume that the input information is certain and cannot naturally represent situations where multiple outcomes are possible.
2) **Dependence on Complete Information:** These approaches often require complete and accurate input data. Missing information can lead to incorrect conclusions.
3) **Difficulty in Handling Noisy Data:** Real-world data may contain errors, measurement variations, or inconsistencies. Fixed rules may produce unreliable results when the input data is noisy.
4) **Limited Flexibility:** Deterministic systems follow predefined rules and may struggle to handle unexpected situations or changing conditions.
5) **Inability to Express Likelihood:** They generally produce definite outputs rather than expressing how likely different outcomes are. Therefore, they cannot naturally distinguish between highly likely and less likely events.

How Probability Reasoning Addresses These Limitations

1) **Handles Uncertainty:** It represents uncertain events using probabilities instead of assuming that every outcome is completely certain.
2) **Works with Incomplete Information:** It estimates the likelihood of possible outcomes using the available evidence, even when some information is missing.
3) **Updates Beliefs with New Evidence:** It allows probabilities to be revised when new information becomes available, improving the assessment of a situation.
4) **Supports Better Decision-Making:** It helps compare different possible outcomes and choose suitable actions based on their likelihood and potential consequences.
5) **Assesses Risks:** It estimates the likelihood of undesirable events, helping individuals and organizations make informed decisions in healthcare, finance, and weather forecasting.

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6) **Handles Variability in Observations:** It provides a mathematical framework for representing uncertainty caused by measurement errors and variations in real-world data.

Example: Medical Diagnosis

1) **Situation:** A patient visits a doctor with fever and cough. These symptoms may be associated with several different diseases, so the exact illness is uncertain.
2) **Deterministic Approach:** A system using fixed rules might associate these symptoms with one particular disease and produce a definite prediction, even though the symptoms alone may not be sufficient for an accurate diagnosis.
3) **Probability Reasoning:** A probabilistic system considers the symptoms, medical history, and available test results to estimate the likelihood of different diseases.
4) **Probability Estimation:** For example, suppose the system estimates a 70% probability of a particular infection based on the available evidence. This is an illustrative value, not an actual clinical estimate.
5) **Use of New Evidence:** If a laboratory test provides new evidence, the system updates the probability of the infection. The doctor can use this updated assessment, along with clinical judgment, to decide whether further testing or treatment is required.
6) **Result:** Unlike a system that produces only a definite outcome, probability reasoning expresses uncertainty and helps the doctor make a better-informed decision.

Thus, probability reasoning overcomes important limitations of deterministic approaches by representing uncertainty, incorporating new evidence, and estimating the likelihood of different outcomes.

Q.20) How does class imbalanced affect on classification? What are the ways to solve class imbalanced problem.

Ans: Class Imbalance and Its Effects on Classification

Class imbalance occurs when the number of samples belonging to one class is significantly greater than the number of samples belonging to another class in a dataset. The class with more samples is called the **majority class**, while the class with fewer samples is called the minority class.

For example, in a fraud detection dataset, 95% of transactions may be genuine and only 5% may be fraudulent. This imbalance can affect the performance of a classification model in the following ways:

1) **Biased Predictions:** The classifier may become biased towards the majority class because it contains more training examples, resulting in poor detection of the minority class.
2) **Misleading Accuracy:** A model may achieve high accuracy by predicting the majority class most of the time, even though it fails to identify important minority-class cases.

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For example, in a fraud detection dataset, 95% of transactions may be genuine and only 5% may be fraudulent. This imbalance can affect the performance of a classification model in the following ways:

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2) **Misleading Accuracy:** A model may achieve high accuracy by predicting the majority class most of the time, even though it fails to identify important minority-class cases.
3) **Poor Minority-Class Detection:** The model may have low recall for the minority class, meaning that it fails to identify many actual minority-class instances.
4) **Difficulty in Learning Patterns:** The model may not receive enough examples of the minority class to learn its characteristics effectively.
5) **Incorrect Decision-Making:** In applications such as fraud detection and disease diagnosis, failure to identify minority-class cases may lead to financial losses or missed diagnoses.
6) **Unreliable Evaluation:** Accuracy alone may not provide a complete picture of model performance. Metrics such as precision, recall, F1-score, and the confusion matrix are also important.

Ways to Solve the Class Imbalance Problem

Several techniques can be used to reduce the effects of class imbalance and improve classification performance.

1) **Oversampling:** Oversampling increases the number of samples belonging to the minority class so that the model receives more examples during training.
   Example: Increasing the number of fraudulent transaction records in a training dataset.
2) **Undersampling:** Undersampling reduces the number of samples belonging to the majority class to make the class distribution more balanced. However, removing too many samples may cause the loss of useful information.
   Example: Reducing the number of genuine transaction records while retaining sufficient data for training.
3) **SMOTE:** SMOTE **(Synthetic Minority Over-sampling Technique)** generates synthetic minority-class samples by interpolating between existing minority-class examples. It helps increase minority-class representation without simply duplicating the same records.
   Example: Creating synthetic fraud examples from similar existing fraud records to improve the model's ability to learn fraud patterns.
4) **Class Weighting:** Class weighting assigns a higher penalty to mistakes involving the minority class during model training. This encourages the classifier to pay more attention to minority-class examples without necessarily changing the dataset size.

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Example: Giving a higher weight to incorrectly classified fraudulent transactions than to incorrectly classified genuine transactions.

5) **Threshold Adjustment:** The classification threshold can be adjusted to change how the model assigns class labels. Lowering the threshold for the positive class may improve recall, although it can also increase false-positive predictions.
Example: Lowering the fraud detection threshold so that more potentially fraudulent transactions are flagged for verification.
6) **Use Suitable Evaluation Metrics:** Instead of relying only on accuracy, evaluate the model using precision, recall, F1-score, and the confusion matrix. Precision-recall curves can also be useful when the positive class is rare.
Example: Evaluating a fraud detection model using recall and F1-score to determine whether it identifies fraudulent transactions effectively.
7) **Collect More Minority-Class Data:** Collecting additional genuine examples of the minority class helps the model learn its patterns more effectively. The new data should be representative and accurately labelled.
Example: Collecting more confirmed fraudulent transaction records to improve fraud detection.
8) **Use Appropriate Algorithms and Validation:** Some algorithms and training methods support class weighting or are better suited to particular imbalanced datasets. Stratified cross-validation helps preserve approximately the same class proportions in each fold, allowing more reliable evaluation.
Example: Using stratified 5-fold cross-validation to evaluate a classifier while maintaining the class distribution across the folds.

Thus, class imbalance can cause biased predictions and poor detection of minority-class instances. Techniques such as oversampling, undersampling, SMOTE, class weighting, threshold adjustment, and suitable evaluation metrics help improve classification performance.

Q.21) Using Mamdani fuzzy model design a fuzzy logic controller to determine the wash time of domestic washing machine. Assume the inputs are dirt and grease on clothes. Use three descriptors for each input variable and five descriptor for output variables. Derive necessary membership functions and required fuzzy rules for the application.

Ans: A Mamdani fuzzy logic controller determines the washing time based on the amount of dirt and grease present on clothes. It uses fuzzy sets, membership functions, and IF–THEN rules to process the inputs. Finally, defuzzification converts the fuzzy output into a specific washing time in minutes. The design of the controller involves the following steps.

Step 1: Identification of Input and Output Variables

The first step is to identify the input and output variables of the washing machine controller. Assume that dirt and grease levels range from 0 to 100, while wash time ranges from 0 to 60 minutes.

- **Input 1 – Dirt Level:** Low, Medium, High.
- **Input 2 – Grease Level:** Low, Medium, High.
- **Output – Wash Time:** Very Short, Short, Medium, Long, Very Long

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- **Input 1 – Dirt Level:** Low, Medium, High.
- **Input 2 – Grease Level:** Low, Medium, High.
- **Output – Wash Time:** Very Short, Short, Medium, Long, Very Long.

Step 2: Derivation of Membership Functions

Membership functions determine the degree to which a value belongs to a fuzzy set. Triangular and trapezoidal membership functions are used to represent the linguistic descriptors of the input and output variables.

i) Dirt Level (Input: 0–100)

- **Low:** Trapezoidal membership function with parameters (0, 0, 20, 40).
- **Medium:** Triangular membership function with parameters (20, 50, 80).
- **High:** Trapezoidal membership function with parameters (60, 80, 100, 100).

ii) B) Grease Level (Input: 0–100)

- **Low:** Trapezoidal membership function with parameters (0, 0, 20, 40).
- **Medium:** Triangular membership function with parameters (20, 50, 80).
- **High:** Trapezoidal membership function with parameters (60, 80, 100, 100).

iii) C) Wash Time (Output: 0–60 minutes)

- **Very Short:** Trapezoidal membership function with parameters (0, 0, 5, 15).
- **Short:** Triangular membership function with parameters (5, 15, 25).
- **Medium:** Triangular membership function with parameters (20, 30, 40).
- **Long:** Triangular membership function with parameters (35, 45, 55).
- **Very Long:** Trapezoidal membership function with parameters (45, 55, 60, 60).

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Membership Function Formulas

For a triangular membership function with parameters (a, b, c):

$$\mu(x) = 0, if\ x \leq a\ or\ x \geq c$$

$$\mu(x) = (x - a) / (b - a), if\ a < x \leq b$$

$$\mu(x) = (c - x) / (c - b), if\ b < x < c$$

$$\mu(x) = 1, if\ x = b$$

For a trapezoidal membership function with parameters (a, b, c, d):

$$\mu(x) = 0, if\ x \leq a\ or\ x \geq d$$

$$\mu(x) = (x - a) / (b - a), if\ a < x < b$$

$$\mu(x) = 1, if\ b \leq x \leq c$$

$$\mu(x) = (d - x) / (d - c), if\ c < x < d$$

For left- and right-shoulder functions, the membership remains 1 at the corresponding extreme.

Step 3: Formulation of Fuzzy Rules

The fuzzy rule base determines the wash time for different combinations of dirt and grease levels. Since each input has three descriptors, a total of nine rules are used to cover all combinations.

1) **Rule 1:** IF dirt is Low AND grease is Low, THEN wash time is Very Short.
2) **Rule 2:** IF dirt is Low AND grease is Medium, THEN wash time is Short.
3) **Rule 3:** IF dirt is Low AND grease is High, THEN wash time is Medium.
4) **Rule 4:** IF dirt is Medium AND grease is Low, THEN wash time is Short.
5) **Rule 5:** IF dirt is Medium AND grease is Medium, THEN wash time is Medium.
6) **Rule 6:** IF dirt is Medium AND grease is High, THEN wash time is Long.
7) **Rule 7:** IF dirt is High AND grease is Low, THEN wash time is Medium.
8) **Rule 8:** IF dirt is High AND grease is Medium, THEN wash time is Long.
9) **Rule 9:** IF dirt is High AND grease is High, THEN wash time is Very Long.

Step 4: Working of the Mamdani Fuzzy Controller

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Step 4: Working of the Mamdani Fuzzy Controller

1) **Fuzzification:** The crisp input values for dirt and grease are converted into membership values using the defined membership functions.
2) **Rule Evaluation:** Each fuzzy rule is evaluated using the AND operator, commonly implemented by taking the minimum of the input membership values.
3) **Aggregation:** The outputs of all activated rules are combined into a single fuzzy output set, commonly using the maximum operator.
4) **Defuzzification:** The aggregated fuzzy output is converted into a crisp wash time using the centroid method.

Formula:

$$Wash\ Time = \int z \times \mu(z)\, dz \,/\, \int \mu(z)\, dz$$

Here, z represents the wash-time value and μ(z) represents the aggregated output membership function. In a discrete implementation, the integrals are replaced by corresponding sums.

Step 5: Example

1) **Input Values:** Suppose the dirt level is 80 and the grease level is 90.
2) **Rule Activation:** Both inputs have high membership, so Rule 9 is strongly activated and recommends a Very Long wash time.
3) **Final Output:** Other rules may also contribute if their membership values are non-zero. After aggregation and centroid defuzzification, the controller produces a specific wash-time value in minutes.

Q.22) Explain the role of representing knowledge in taxonomies and ontologies, and how advanced analytics can be applied to cognitive systems.

Ans:

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1) Knowledge representation is the process of organizing information in a structured form so that a computer system can understand relationships between concepts and use them for reasoning and decision-making.

2) Taxonomies and ontologies help cognitive systems organize knowledge, while advanced analytics helps them discover patterns, generate predictions, and produce useful insights from large amounts of data.

Role of Taxonomies in Knowledge Representation

A taxonomy is a hierarchical classification system that organizes concepts into categories and subcategories based on their relationships. It helps cognitive systems classify information and retrieve relevant knowledge efficiently.

1) **Hierarchical Organization:** Taxonomies arrange concepts from general categories to specific subcategories, making information easier to understand and manage.

2) **Classification of Information:** They help cognitive systems classify documents, objects, and data into appropriate categories.

3) **Improved Information Retrieval:** Taxonomies help users and systems locate relevant information by organizing related concepts under common categories.

4) **Consistency:** They provide a standardized structure for classifying information across different systems and applications.

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5) **Example:** In a healthcare taxonomy, the category *Diseases* may contain subcategories such as *Infectious Diseases* and *Cardiovascular Diseases*.

Role of Ontologies in Knowledge Representation

An ontology is a formal representation of concepts, properties, relationships, and rules within a particular domain. Unlike a taxonomy, which mainly organizes concepts hierarchically, an ontology represents richer relationships between concepts and supports logical reasoning.

1) **Representation of Concepts:** Ontologies define important entities and concepts within a domain, such as patients, diseases, medicines, and treatments.

2) **Relationship Identification:** They describe relationships between concepts. For example, a patient may have a disease, and a medicine may be used to treat that disease.

3) **Knowledge Sharing:** Ontologies provide a common vocabulary that enables different systems to exchange and interpret information consistently.

4) **Reasoning and Inference:** Cognitive systems can use ontology relationships and rules to infer additional information from existing knowledge.

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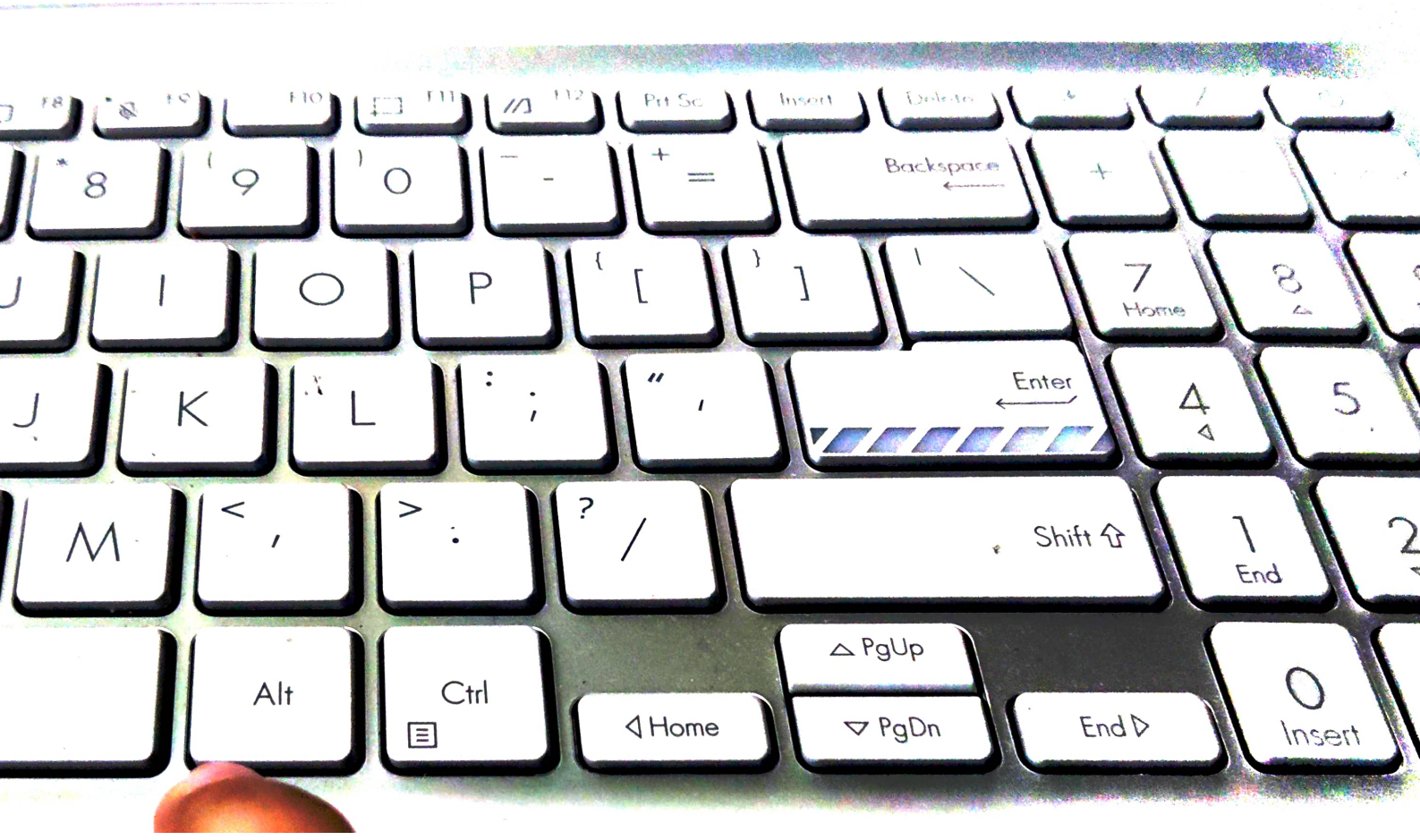
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5) **Knowledge Integration:** Ontologies help combine information from different sources by providing a common structure and meaning.

6) **Example:** In a medical ontology, if a particular medicine is linked to the treatment of a disease, a cognitive system can use this relationship to retrieve relevant treatment information.

Application of Advanced Analytics in Cognitive Systems

Advanced analytics uses statistical techniques, machine learning, data mining, and predictive modelling to extract meaningful insights from data. In cognitive systems, it helps the system learn from information, identify patterns, make predictions, and support intelligent decisions.

1) **Predictive Analytics:** It uses historical data and machine learning models to predict future events or outcomes.

   Example: Predicting the risk of a disease using patient records.

2) **Descriptive Analytics:** It summarizes historical data to identify trends, patterns, and relationships.

   Example: Analyzing customer purchasing behaviour to understand buying patterns.

3) **Prescriptive Analytics:** It recommends suitable actions based on predictions, objectives, and constraints.

   Example: Recommending an appropriate treatment option for a doctor to evaluate.

4) **Natural Language Processing (NLP):** It enables cognitive systems to analyze and understand human language from documents, messages, and conversations.

   Example: A virtual assistant understanding a user's question and providing a relevant response.

5) **Pattern Recognition:** Machine learning algorithms identify recurring patterns and relationships in large datasets.

   Example: Detecting unusual financial transactions that may indicate fraud.

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6) **Decision Support:** Advanced analytics combines data, rules, and predictive results to help users make informed decisions.

   Example: A business intelligence system analyzing sales trends to support inventory planning.

7) **Continuous Learning:** Cognitive systems can use new data and feedback to update models and improve future predictions, depending on their design.

   Example: A recommendation system adapting its suggestions according to a user's recent viewing history.

Integration of Taxonomies, Ontologies, and Advanced Analytics

1) Taxonomies organize information into categories and subcategories.
2) Ontologies define concepts and the relationships between them.
3) Advanced analytics processes structured and unstructured data to discover patterns and generate predictions.
4) The cognitive system combines the organized knowledge and analytical results to support reasoning, recommendations, and decision-making.

Thus, taxonomies and ontologies provide structured and meaningful knowledge, while advanced analytics enables cognitive systems to analyze data, identify patterns, make predictions, and support intelligent decision-making.

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Q.23) Discuss the role of utility and expected utility in decision theory with example.

Ans:

1) Decision theory is a branch of decision-making that helps individuals or systems choose the best action from available alternatives under conditions of certainty, risk, or uncertainty.
2) **Utility** measures the satisfaction or value associated with an outcome, while **expected utility** calculates the average utility of possible outcomes by considering their probabilities.

Role of Utility in Decision Theory

Utility represents the preference, satisfaction, or benefit that a decision-maker receives from a particular outcome. It helps compare different choices according to how desirable their results are.

1) **Measuring Preferences:** Utility assigns a value to an outcome based on how desirable it is to the decision-maker. A higher utility generally represents a more preferred outcome.
2) **Comparing Alternatives:** It helps compare different actions and select the one that provides greater benefit or satisfaction.
3) **Representing Risk Attitudes:** A utility function can represent whether a decision-maker is risk-averse, risk-neutral, or

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2) **Comparing Alternatives:** It helps compare different actions and select the one that provides greater benefit or satisfaction.
3) **Representing Risk Attitudes:** A utility function can represent whether a decision-maker is risk-averse, risk-neutral, or risk-seeking.
4) **Supporting Rational Decisions:** Utility provides a systematic basis for selecting actions according to the decision-maker's preferences and objectives.
5) **Modelling Satisfaction:** Utility helps represent the level of satisfaction or benefit a person receives from a particular result.
6) **Assigning Values to Outcomes:** It assigns numerical values to possible outcomes, making it easier to evaluate and compare their desirability.

Role of Expected Utility in Decision Theory

Expected utility is the probability-weighted average of the utilities of all possible outcomes of a decision. It is particularly useful when the result is uncertain and different outcomes have different probabilities.

The formula for expected utility is:

$$EU(A) = \Sigma P(O_i \mid A) \times U(O_i)$$

Where:

- **EU(A)** = Expected utility of action A
- **P(O_i | A)** = Probability of outcome O_i when action A is chosen
- **U(O_i)** = Utility of outcome O_i
- **Σ** = Sum over all possible outcomes

1) **Handling Uncertainty:** Expected utility helps evaluate decisions when the exact outcome is not known in advance.
2) **Considering Probabilities:** It considers both the utility of each possible outcome and the probability of that outcome occurring.

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3) **Selecting the Best Action:** A decision-maker can compare the expected utilities of available actions and choose the action with the highest expected utility, assuming the utility function correctly represents their preferences.

4) **Evaluating Risk:** It helps decision-makers compare uncertain options and understand the trade-off between possible benefits and losses.

5) **Supporting Decision-Making Under Risk:** It provides a systematic method for evaluating alternatives when the possible outcomes and their probabilities are known or can be estimated.

6) **Comparing Uncertain Choices:** It allows decision-makers to compare options that have different possible outcomes and probabilities using a common measure.

7) **Improving Resource Allocation:** Expected utility can help allocate money, time, or other resources toward alternatives that offer the greatest expected benefit according to the decision-maker's preferences.

Example: Choosing an Investment

Suppose a person has two investment options, A and B. The possible outcomes, their probabilities, and their utility values are given below.

Investment	Outcome	Probability	Utility
A	High return	0.7	100
A	Low return	0.3	20
B	High return	0.4	150
B	Low return	0.6	40

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Step 1: Calculate the Expected Utility of Investment A

$$EU(A) = (0.7 \times 100) + (0.3 \times 20)$$

$$EU(A) = 70 + 6$$

$$\boldsymbol{EU(A) = 76}$$

Step 2: Calculate the Expected Utility of Investment B

$$EU(B) = (0.4 \times 150) + (0.6 \times 40)$$

$$EU(B) = 60 + 24$$

$$\boldsymbol{EU(B) = 84}$$

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Step 3: Compare the Results

The expected utility of Investment A is 76, whereas the expected utility of Investment B is 84. Since Investment B has the higher expected utility, the decision-maker would choose Investment B, provided that the given utility values accurately represent their preferences.

Thus, utility helps measure the desirability of outcomes, while expected utility combines these values with their probabilities to support rational decision-making under uncertainty.

Q.23) Differentiate between the following:

i) RNN and ANN

Ans:

Sr. No.	Basis	ANN (Artificial Neural Network)	RNN (Recurrent Neural Network)
1)	**Definition**	ANN is a neural network that processes information through interconnected neurons arranged in layers.	RNN is a neural network designed to process sequential data by using connections that maintain information from previous steps.
2)	**Architecture**	It generally consists of input, hidden, and output layers with feedforward connections.	It contains recurrent connections that allow information from previous time steps to influence the current output.
3)	**Memory**	A basic feedforward ANN does not maintain a hidden state across sequential inputs.	RNN maintains a hidden state that carries information from earlier inputs in a sequence.

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4)	**Data Processing**	It is commonly used for independent inputs where the order of observations is not important.	It is suitable for sequential data where the order of observations matters.
5)	**Input Handling**	It typically processes fixed-size input features for each prediction.	It can process sequences of varying lengths, depending on its implementation.
6)	**Learning**	It learns patterns by adjusting weights and biases during training.	It learns sequential patterns by updating weights across time steps, typically using backpropagation through time.
7)	**Complexity**	A basic feedforward ANN is generally simpler to train because it does not require recurrent computation across time steps.	RNN training is generally more complex because information and gradients are propagated through sequences.
8)	**Applications**	It is used for classification, regression, pattern recognition, and predicting outcomes from structured data.	It is used for language modelling, speech recognition, time-series forecasting, and sequence prediction.
9)	**Examples**	Predicting house prices based on area, location, and number of rooms.	Predicting the next word in a sentence based on the words that came before it.
10)	**Limitations**	A basic feedforward ANN does not naturally capture dependencies between successive time steps.	A standard RNN may suffer from vanishing or exploding gradients when learning long-term dependencies.

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i) Fuzzyness and probability

Ans:

Sr. No.	Basis	Fuzziness (Fuzzy Logic)	Probability
1)	**Meaning**	Fuzziness represents the degree to which an object belongs to a particular category.	Probability represents the likelihood that an event will occur.
2)	**Value Range**	The membership value ranges from 0 to 1.	The probability of an event ranges from 0 to 1.
3)	**Type of Uncertainty**	It handles vagueness and imprecise information.	It handles uncertainty about whether an event will occur.

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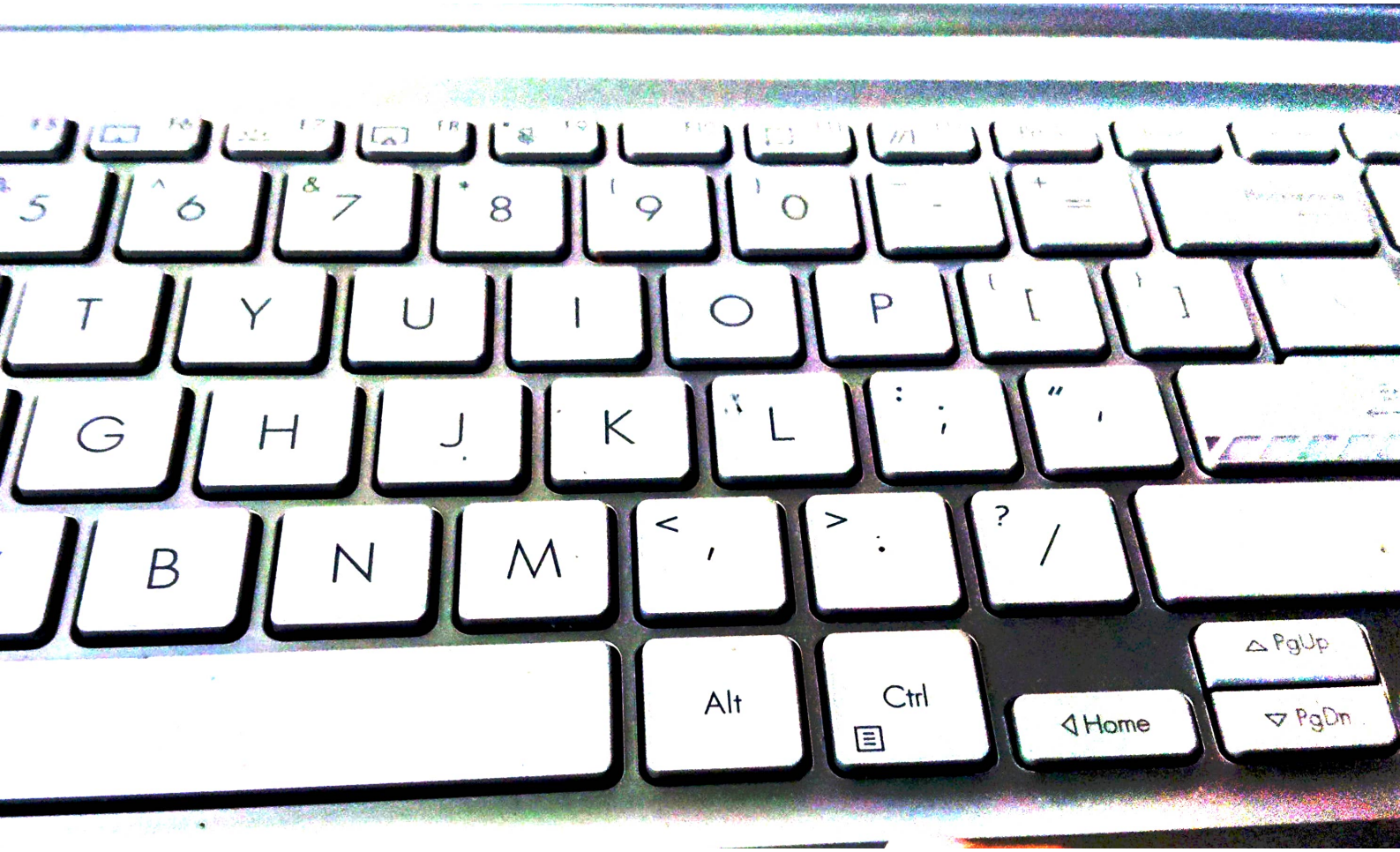
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ii) Fuzzyness and probability

Ans:

Sr. No.	Basis	Fuzziness (Fuzzy Logic)	Probability
1)	**Meaning**	Fuzziness represents the degree to which an object belongs to a particular category.	Probability represents the likelihood that an event will occur.
2)	**Value Range**	The membership value ranges from 0 to 1.	The probability of an event ranges from 0 to 1.
3)	**Type of Uncertainty**	It handles vagueness and imprecise information.	It handles uncertainty about whether an event will occur.
4)	**Interpretation of 0.5**	A membership value of 0.5 indicates partial membership in a fuzzy set.	A probability of 0.5 indicates a 50% chance of an event occurring.
5)	**Basis**	It is based on membership functions and degrees of belonging.	It is based on possible outcomes, observations, or probability models.
6)	**Example**	A temperature of 30°C may belong to the fuzzy category "Hot" with a membership value of 0.7.	The probability of rain tomorrow may be 0.7, indicating a 70% chance of rain.
7)	**Nature**	It describes how strongly an object satisfies a vague concept.	It describes how likely an event is to occur.
8)	**Mathematical Representation**	It uses a membership function, written as μA(x), to represent the degree of membership of x in fuzzy set A.	It uses a probability function, written as P(A), to represent the probability of event A.
9)	**Applications**	It is used in fuzzy controllers, decision systems, and systems involving vague concepts such as "hot" or "slow."	It is used in weather forecasting, risk analysis, medical diagnosis, and statistical prediction.
10)	**Main Objective**	Its objective is to represent and process vague or imprecise concepts.	Its objective is to quantify uncertainty about events and their possible outcomes.

iii) Crips set and fuzzy set

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Sr. No.	Basis	Crisp Set	Fuzzy Set
1)	**Definition**	An element either belongs to the set or does not belong to it.	An element can belong to the set partially or completely.
2)	**Membership Value**	Membership value is either 0 or 1.	Membership value can be any value between 0 and 1, inclusive.
3)	**Boundary**	It has clearly defined boundaries between membership and non-membership.	Its boundaries can be gradual rather than sharply defined.
4)	**Logic Used**	It is based on classical Boolean logic.	It is based on fuzzy logic.
5)	**Membership Function**	The membership function assigns only 0 or 1 to an element.	The membership function assigns a degree of membership between 0 and 1.
6)	**Nature of Information**	It is suitable for clearly defined and precise categories.	It is suitable for vague or imprecise concepts.
7)	**Example**	A person aged 18 or above belongs to the set of adults if adulthood is defined as age ≥ 18.	A temperature of 30°C may belong to the fuzzy set "Hot" with a membership value of 0.7.
8)	**Decision-Making**	Decisions are made using definite membership conditions.	Decisions can consider different degrees of membership.
9)	**Applications**	It is used in database queries, digital circuits, and binary classification rules.	It is used in fuzzy controllers, intelligent systems, and decision-making involving vague concepts.
10)	**Flexibility**	It is less flexible when representing gradual changes between categories.	It can represent gradual transitions between categories more naturally.

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iv) Bagging and boosting

Ans:

Sr. No.	Basis	Bagging	Boosting

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v) Bagging and boosting

Ans:

Sr. No.	Basis	Bagging	Boosting
1)	**Meaning**	Bagging stands for Bootstrap Aggregating and combines predictions from multiple models trained on different bootstrap samples.	Boosting combines multiple weak learners sequentially to build a stronger predictive model.
2)	**Training Method**	Models are trained independently and can generally be trained in parallel.	Models are trained sequentially, with each model influenced by the performance of previous models.
3)	**Main Objective**	It mainly reduces variance and helps prevent overfitting in unstable models.	It mainly reduces bias by improving the model's ability to learn difficult patterns.
4)	**Data Sampling**	It commonly uses bootstrap samples created by randomly sampling the training data with replacement.	It commonly adjusts sample weights or fits subsequent models to residuals or errors, depending on the boosting algorithm.
5)	**Error Handling**	Models are trained independently, without specifically focusing each new model on previous models' errors.	Each successive model attempts to correct errors or improve upon the predictions of earlier models.
6)	**Combining Predictions**	Predictions are combined using majority voting for classification or averaging for regression.	Predictions are combined using a weighted sum or another algorithm-specific combination method.
7)	**Model Dependency**	Each model is generally independent of the other models during training.	Each successive model depends on information from earlier models.
8)	**Overfitting**	It often reduces overfitting by reducing variance, especially when individual models are unstable.	It can overfit noisy data if not properly controlled, although regularization and other techniques can reduce this risk.
9)	**Examples**	Random Forest is a common example of a bagging-based ensemble method.	AdaBoost, Gradient Boosting, and XGBoost are examples of boosting methods.

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10)	**Applications**	It is used in classification, regression, and prediction tasks where reducing model variance is important.	It is used in classification, regression, fraud detection, and ranking tasks where improving predictive accuracy is important.

Q.24) Write a short note on:

i) Trends in Data science for audio

Ans: Data science for audio involves analyzing and processing audio signals using machine learning, deep learning, and artificial intelligence. It is widely used in speech recognition, music analysis, voice assistants, and sound classification. The major trends in data science for audio are as follows:

1) **Automatic Speech Recognition (ASR):** It converts spoken language into written text using machine learning and deep learning techniques. It is used in voice typing, transcription, and virtual assistants.
2) **Natural Language Processing for Speech:** It combines audio processing with language understanding to interpret spoken commands and conversations. It helps systems understand user requests and generate suitable responses.
3) **Voice Assistants and Conversational AI:** Audio data science enables systems such as voice assistants and conversational agents to recognize speech and respond naturally to users.
4) **Emotion and Sentiment Recognition:** Machine learning models analyze voice characteristics such as pitch, tone, and speaking speed to identify possible emotional states in speech.
5) **Music Analysis and Recommendation:** Data science techniques analyze rhythm, melody, tempo, and genre to classify music and recommend songs based on user preferences.
6) **Audio Classification and Sound Detection:** Deep learning models identify different sounds, such as traffic noise, alarms, animal sounds, and industrial machine sounds. This is useful in surveillance and environmental monitoring.
7) **Speech Synthesis and Voice Cloning:** AI models generate artificial speech from text and can reproduce voice characteristics. These technologies are used in text-to-speech systems, accessibility tools, and digital assistants.
8) **Noise Reduction and Audio Enhancement:** Machine learning techniques remove unwanted background noise and improve speech clarity in recordings, online meetings, and communication systems.
9) **Multimodal AI:** Modern AI systems combine audio with text, images, or video to understand information more effectively. For example, a system can use speech and visual information together to understand a video.
10) **Real-Time Audio Analytics:** AI systems process audio as it is received, enabling applications such as live transcription, call analysis, and immediate detection of important sounds.

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analysis, and immediate detection of important sounds.

Thus, data science for audio is advancing through AI-based speech recognition, sound analysis, audio generation, and real-time processing, making audio systems more intelligent and useful across various industries.

ii) Mc-Culloch pits neuron with an example

Ans: The McCulloch-Pitts (M-P) neuron is one of the earliest mathematical models of an artificial neuron, proposed by Warren McCulloch and Walter Pitts in 1943. It is a binary threshold model that accepts binary inputs, processes them using weights and a threshold, and produces a binary output of either 0 or 1.

Working of the McCulloch-Pitts Neuron

1) **Inputs:** The neuron receives binary inputs such as x_1 and x_2, where each input can be either 0 or 1.
2) **Weights:** Each input is multiplied by its corresponding weight (w_1, w_2). The weights determine the contribution of each input to the total sum.
3) **Summation:** The neuron calculates the weighted sum of its inputs.

$$\boldsymbol{Net\ Input} = \boldsymbol{x_1 w_1 + x_2 w_2}$$

4) **Threshold Function:** The weighted sum is compared with a fixed threshold (θ). If the sum is greater than or equal to the threshold, the output is 1; otherwise, it is 0.

$$\boldsymbol{y = 1, if\ Net\ Input \geq \theta}$$

$$\boldsymbol{y = 0, if\ Net\ Input < \theta}$$

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Example: AND Gate Using an M-P Neuron

An AND gate produces output 1 only when both inputs are 1. Consider an M-P neuron with weights $w_1 = 1$, $w_2 = 1$ and threshold $\theta = 2$.

The output is calculated as follows:

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Input x_1	Input x_2	Net Input ($x_1 + x_2$)	Output y
0	0	0	0
0	1	1	0
1	0	1	0
1	1	2	1

Explanation:

1) When both inputs are 0, the net input is 0, which is below the threshold. Therefore, the output is 0.
2) When one input is 1 and the other is 0, the net input is 1, which is below the threshold. Therefore, the output is 0.
3) When both inputs are 1, the net input is 2, which equals the threshold. Therefore, the output is 1.

Thus, the McCulloch-Pitts neuron implements the AND logic gate using binary inputs, fixed weights, and a threshold function.

iii) Describe NLP in support of cognitive system

Ans: Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that enables computers to understand, interpret, process, and generate human language. In cognitive systems, NLP helps machines interact with humans using natural languages such as English, Hindi, and Marathi. It enables cognitive systems to analyze text and speech, extract meaningful information, and support intelligent decision-making.

Role of NLP in Supporting Cognitive Systems

1) **Understanding Human Language:** NLP enables cognitive systems to understand user queries, commands, and conversations by analyzing the meaning of words and sentences.
2) **Text Analysis:** It processes large amounts of textual data, such as documents, emails, reports, and articles, to extract useful information.
3) **Speech Recognition:** NLP works with speech recognition technologies to convert spoken language into text, allowing users to interact with cognitive systems through voice commands.
4) **Sentiment Analysis:** It identifies opinions, emotions, and attitudes expressed in text. For example, a system can analyze customer reviews to determine whether feedback is positive or negative.
5) **Information Extraction:** NLP identifies important details such as names, dates, locations, and key facts from unstructured text.

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6) **Language Translation:** It helps cognitive systems translate text from one language to another, supporting communication between people who speak different languages.

7) **Question Answering:** NLP enables cognitive systems to interpret questions and provide relevant answers using available information and knowledge.

8) **Chatbots and Virtual Assistants:** NLP helps chatbots and virtual assistants understand user requests and generate appropriate responses, improving human-computer interaction.

9) **Text Summarization:** It helps cognitive systems create shorter versions of lengthy documents while preserving the main ideas and important information.

10) **Decision Support:** NLP extracts meaningful insights from textual data and presents relevant information to help users make informed decisions.

Example: Healthcare Cognitive System

A healthcare cognitive system can use NLP to analyze a patient's medical records, understand symptoms described in natural language, extract important medical information, and provide relevant information to assist a doctor. The doctor makes the final medical decision.

Thus, NLP supports cognitive systems by enabling them to understand human language, extract knowledge, communicate naturally, and assist in intelligent decision-making.

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iv) **Bootstrap for sampling**

Ans: Bootstrap sampling is a statistical resampling technique used to estimate the accuracy, variability, and reliability of a sample statistic. It involves repeatedly selecting samples from the original dataset with replacement, meaning the same observation can be selected multiple times. It is widely used in data science and machine learning to estimate confidence intervals and evaluate model performance.

Steps Involved in Bootstrap Sampling

1) **Original Dataset:** Start with an available dataset containing N observations.

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2) **Resampling with Replacement:** Randomly select N observations from the original dataset with replacement. Some observations may appear multiple times, while others may not appear at all.

3) **Generate Bootstrap Samples:** Repeat the resampling process many times to create multiple bootstrap datasets of the same size as the original dataset.

4) **Calculate Statistics:** Calculate the required statistic, such as the mean, median, variance, or model accuracy, for each bootstrap sample.

5) **Estimate Variability:** Compare the calculated statistics across bootstrap samples to understand how much the statistic varies.

6) **Calculate Confidence Intervals:** Use the distribution of bootstrap estimates to calculate a confidence interval for the population statistic.

Example: Consider an original dataset containing five values: **[10, 20, 30, 40, 50]**.

Two possible bootstrap samples of the same size are:

- **Sample 1:** [10, 20, 20, 40, 50]
- **Sample 2:** [10, 30, 30, 40, 40]

In Sample 1, the value 20 appears twice, while 30 is not selected. In Sample 2, the values 30 and 40 appear twice. This is possible because sampling is performed with replacement.

The mean of Sample 1 is:

$$\boldsymbol{Mean} = (10 + 20 + 20 + 40 + 50) / 5$$

$$\boldsymbol{Mean} = \mathbf{28}$$

The mean of Sample 2 is:

$$\boldsymbol{Mean} = (10 + 30 + 30 + 40 + 40) / 5$$

$$\boldsymbol{Mean} = \mathbf{30}$$

By generating many such bootstrap samples and calculating their means, we can estimate the variability and reliability of the original sample mean.

Thus, bootstrap sampling estimates the uncertainty of statistical results by repeatedly resampling the original dataset with replacement.

v) **Holdout method and Random subsampling**

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v) **Holdout method and Random subsampling**

Ans: The holdout method and random subsampling are data sampling techniques used to evaluate the performance of machine learning models. They divide a dataset into training and testing sets to check how well a model performs on unseen data.

Holdout Method

The holdout method is a model evaluation technique in which a dataset is divided into two separate parts: a training set and a testing set. The model learns from the training set, and its performance is evaluated using the testing set.

1) **Data Splitting:** The dataset is divided into training and testing sets, such as 80% for training and 20% for testing.
2) **Model Training:** The training set is used to train the machine learning model and learn patterns from the data.
3) **Model Testing:** The testing set is used to evaluate the trained model on data that was not used during training.
4) **Performance Evaluation:** Evaluation metrics such as accuracy, precision, recall, and F1-score are used to measure the model's performance.
5) **Advantages:** The holdout method is simple, easy to implement, and computationally efficient because the model is trained only once.
6) **Limitation:** The result may depend on the particular train-test split. If the split is not representative, the estimated performance may be unreliable.

Example: Suppose a dataset contains 1,000 student records. Using an 80:20 split, 800 records are used to train a model that predicts student performance, while the remaining 200 records are used to test its accuracy.

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Random Subsampling

Random subsampling is a model evaluation technique in which the dataset is randomly divided into training and testing sets multiple times. The model is trained and tested repeatedly using different splits, and the results are combined to estimate its overall performance.

1) **Random Data Splitting:** The dataset is randomly divided into training and testing sets for each iteration.

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2) **Repeated Training and Testing:** The model is trained on the training set and evaluated on the testing set in every iteration.

3) **Different Samples:** Each iteration uses a different random split. The same observation may appear in a training set in one iteration and a testing set in another.

4) **Average Performance:** The performance scores from all iterations are averaged to obtain a more stable estimate of model performance.

5) **Advantages:** It reduces dependence on a single random split and provides a better understanding of how the model performs across different samples.

6) **Limitation:** It requires more computational time because the model is trained and tested multiple times. Some observations may also be selected for testing more frequently than others.

Example: Suppose a dataset contains 1,000 student records. The dataset is randomly split into 800 training records and 200 testing records five times. The model is trained and evaluated in each iteration, and the five accuracy scores are averaged to estimate its performance.

Thus, the holdout method evaluates a model using one train-test split, whereas random subsampling repeats the splitting and evaluation process to obtain an average performance estimate.

vi) Significance of ROC curves

Ans: A Receiver Operating Characteristic (ROC) curve is a graphical representation used to evaluate the performance of a binary classification model at different classification thresholds. It plots the True Positive Rate (TPR) against the False Positive Rate (FPR) to show how effectively the model distinguishes between two classes.

Significance of ROC Curves

1) **Evaluates Classifier Performance:** The ROC curve helps assess how well a classification model distinguishes between positive and negative classes.

2) **Shows True Positive Rate:** The True Positive Rate (TPR), also called sensitivity or recall, measures the proportion of actual positive cases correctly identified by the model.

$$TPR = TP / (TP + FN)$$

3) **Shows False Positive Rate:** The False Positive Rate (FPR) measures the proportion of actual negative cases incorrectly classified as positive.

$$FPR = FP / (FP + TN)$$

4) **Compares Classification Models:** ROC curves allow comparison of different classification models. A curve closer to the top-left corner generally indicates better classification performance.

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$$FPR = FP / (FP + TN)$$

4) **Compares Classification Models:** ROC curves allow comparison of different classification models. A curve closer to the top-left corner generally indicates better classification performance.

5) **Helps Select Classification Thresholds:** The curve shows how changing the decision threshold affects true positives and false positives, helping select a suitable threshold for a particular application.

6) **Measures Performance Using AUC:** The Area Under the ROC Curve (AUC) summarizes the model's ability to distinguish between the two classes. An AUC of 1.0 indicates perfect discrimination, while an AUC of 0.5 indicates performance equivalent to random ranking.

7) **Useful for Different Applications:** ROC curves are used in medical diagnosis, fraud detection, spam filtering, and other binary classification tasks where the consequences of false positives and false negatives matter.

Example: In a medical diagnosis system, an ROC curve can help evaluate how effectively a model identifies patients who have a disease while limiting false alarms among healthy patients. The threshold can be adjusted depending on whether missing a disease or incorrectly identifying a healthy person is more costly.

Thus, ROC curves help evaluate, compare, and tune binary classification models by illustrating the trade-off between correctly detecting positive cases and incorrectly classifying negative cases.

Numerical

Q.1) Calculate Accuracy, precision, recall, sensitivity and specificity for the following example.

Q.2) Implement AND, XOR function using Mc-Culloch Pitts model.

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MUSA (MUMBAI UNIVERSITY STUDENTS ASSOCIATION)

TOP 15 QUESTION BANK OF AI&DS-II

B.E SEM-VII **SCHEME: R-19 C SCHEME** **FOR WINTER SESSION 2026**

BRANCH: IT **BATCH: REGULAR & KT**

Q.1) What is the Bayesian Belief Network? Explain Interfacing in detail with example.

Q.2) Explain the architecture and components of CNN in detail and state its application also.

Q.3) What is a Markov Decision Process (MDP)? List its primary components and need of MDP in detail.

Q.4) Explain the architecture and key components of ANN in details.

Q.5) Define Defuzzification. Discuss any two methods of defuzzification?

Q.6) Define Cognitive computing. Draw a neat diagram of components of the cognitive system and explain the components also List and explain the design principle of Cognitive system.

Q.7) Explain different types of membership functions used in fuzzy logic with suitable example. Describe the Properties of Fuzzy Sets with an example.

Q.8) What are all the objects of using LSTEM? Explain its architecture.

Q.9) Explain various metrics for evaluating the classifier performance. Discuss any two cross validation methods.

Q.10) Briefly explain Data science for Multimodal applications.

Q.11) What is cognitive computing, and how does it differ from traditional computing? List and explain the process of building a cognitive application.

Q.12) How bagging and boosting handle bias-variance trade-off differently, and analyze their

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Q.9) Explain various metrics for evaluating the classifier performance. Discuss any two cross validation methods.

Q.10) Briefly explain Data science for Multimodal applications.

Q.11) What is cognitive computing, and how does it differ from traditional computing? List and explain the process of building a cognitive application.

Q.12) How bagging and boosting handle bias-variance trade-off differently, and analyze their effectiveness in dealing with noisy data and overfitting. Explain with algorithmic such as Random forest and AdaBoost.

Q.13) Using Mamdani fuzzy model design a fuzzy logic controller to determine the wash time of domestic washing machine. Assume the inputs are dirt and grease on clothes. Use three descriptors for each input variable and five descriptor for output variables. Derive necessary membership functions and required fuzzy rules for the application.

Q.14) Explain the role of representing knowledge in taxonomies and ontologies, and how advanced analytics can be applied to cognitive systems.

Q.15) Discuss the role of utility and expected utility in decision theory with example.

Q.16) Do all the differentiate between, short note and numerical from question bank.

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